Ultrafast Transient-Grating Spectroscopy of GaAs

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Abstract

In this experiment, the relaxation and diffusion of photoexcited carriers in GaAs are studied using pump–probe and transient-grating measurements. First, the laser pulses are characterized by their spectrum and interferometric autocorrelation. The fitted pulse durations decrease from (112.89 ± 0.10) fs before compression to (93.902 ± 0.054) fs afterwards, while the spectrum gives a Gaussian transform limit of (83.87 ± 0.24) fs. These uncertainties are formal standard errors; the dependence on the fitting interval is larger. We then fit the carrier signals with two exponential decays. From the dependence of the slow transient-grating rate on the grating period, we obtain an effective diffusion coefficient of $(3.71 \pm 0.70) \times 10^{-3}\,\mathrm{m^2\,s^{-1}}$, with the uncertainty given by the regression standard error. However, the pump–probe decay does not agree with the extrapolation of these rates. The fitted diffusion coefficient therefore depends on the assumed assignment of the slow component, and a common recombination lifetime cannot be established. The available measurements also do not determine an electron–phonon coupling coefficient.

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1 Introduction

When a semiconductor absorbs a short laser pulse, electrons are excited and leave holes behind. The excited carriers lose energy, recombine, and move through the sample. These processes change its optical properties, so a second, delayed pulse can be used to follow the response. In a pump–probe (PP) measurement, we measure the change in reflected probe light. In transient-grating (TG) spectroscopy, two pump beams instead create a periodic excitation pattern. Carriers diffuse from regions of high density into regions of lower density, reducing the contrast of this pattern. Measuring how quickly gratings of different periods disappear allows us to study carrier transport.

In this experiment, we study these processes in GaAs. Before analyzing the carrier dynamics, we determine the pulse duration and spectral bandwidth to characterize the excitation. The analysis then follows the four parts of the tutorial [3]: pulse characterization, hot-carrier cooling, transient-reflectivity and grating decay times, and the determination of transport quantities. We compare the results with the expected diffusion dependence and discuss where the measurements are insufficient to assign a fitted decay to a particular physical process.

2 Theoretical background

2.1 Transient-grating formation and carrier decay

An ultrashort pump pulse creates electron–hole pairs in GaAs. Two pump fields with equal intensity I_0 and wave vectors $\mathbf{k}_1, \mathbf{k}_2$ produce the intensity pattern

$$I(\mathbf{r}) = |E_1 + E_2|^2 \tag{1}$$

$$= |E_1|^2 + |E_2|^2 + E_1 E_2^* + E_1^* E_2 \tag{2}$$

$$= 2I_0 \left[1 + \cos\left((\mathbf{k}_1 - \mathbf{k}_2) \cdot \mathbf{r} + \phi\right)\right]. \tag{3}$$

The distance between neighboring excitation maxima is the grating period Λ_{TG}. We describe this spatial variation by the wave number $q = 2\pi/\Lambda_{\mathrm{TG}}$. A smaller period therefore gives a larger q. At the same modulation amplitude, it also gives a steeper density gradient, so diffusion reduces the grating more quickly. The wave number describes variation in space, while the optical frequency describes oscillation in time. For an angle θ between the pump beams, $q = |\mathbf{k}_1 - \mathbf{k}_2| = 2k\sin(\theta/2)$, giving

$$\Lambda_{\mathrm{TG}} = \frac{2\pi}{q} = \frac{\lambda}{2\sin(\theta/2)}. \tag{4}$$

The two-beam interference and resulting spatial carrier modulation are illustrated in Fig. 1.

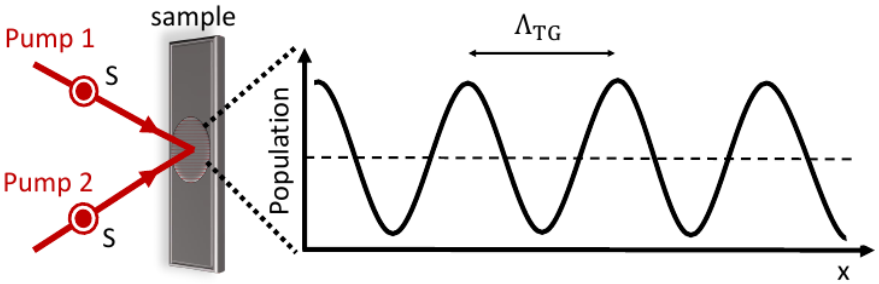


Figure 1: Two crossed pump beams create a sinusoidal excitation pattern in the sample. The grating period Λ_{TG} sets the spatial wave number $q = 2\pi/\Lambda_{\mathrm{TG}}$ used in the diffusion analysis. Reproduced from Fig. 4.5 of the course tutorial.

Write the carrier density as a uniform part plus its sinusoidal grating component,

$$n(x,t) = N_0(t) + N_q(t)\cos(qx). \tag{5}$$

Substitution into the tutorial's one-dimensional diffusion–recombination equation gives

$$\frac{\partial n(x,t)}{\partial t} = D_a \frac{\partial^2 n(x,t)}{\partial x^2} - \frac{n(x,t)}{\tau_{\rm rec}}. \tag{6}$$

Using $\partial_t n = \dot{N}_0 + \dot{N}_q \cos(qx)$ and $\partial_x^2 n = -q^2 N_q \cos(qx)$, and equating the coefficients of 1 and $\cos(qx)$, respectively,

$$\dot{N}_0 = -\frac{N_0}{\tau_{\rm rec}}, \qquad \dot{N}_q = -\left(D_a q^2 + \frac{1}{\tau_{\rm rec}}\right) N_q. \tag{7}$$

$$\int_{N_0(0)}^{N_0(t)} \frac{dN_0}{N_0} = -\int_0^t \frac{dt'}{\tau_{\rm rec}}, \qquad \int_{N_q(0)}^{N_q(t)} \frac{dN_q}{N_q} = -\int_0^t \left(D_a q^2 + \frac{1}{\tau_{\rm rec}}\right) dt'. \tag{8}$$

Evaluation yields

$$N_0(t) = N_0(0) e^{-t/\tau_{\rm rec}}, \tag{9}$$

$$N_q(t) = N_q(0) e^{-(D_a q^2 + 1/\tau_{\rm rec})t}. \tag{10}$$

For heterodyne detection with a constant local oscillator and $E_{\rm dif} = \alpha N_q$,

$$|E_{\rm LO} + E_{\rm dif}|^2 = |E_{\rm LO}|^2 + |\alpha|^2 N_q^2 + 2\,{\rm Re}(E_{\rm LO}^* \alpha) N_q, \tag{11}$$

$$S_{\rm het} \propto N_q \propto e^{-(D_a q^2 + 1/\tau_{\rm rec})t}. \tag{12}$$

Thus the heterodyne decay rate contains $D_a q^2$, with no factor of two from squaring the grating amplitude. Assuming that the reflectivity change is also linear in the corresponding population, the $q = 0$ transient-reflectivity and finite-q grating traces therefore have the forms

$$S_{\rm pop}(t) = A_{\rm pop} e^{-t/\tau_{\rm rec}} + C_{\rm pop}, \tag{13}$$

$$S_{\rm TG}(t) = A_{\rm TG} e^{-t/\tau_{\rm TG}} + C_{\rm TG}, \tag{14}$$

$$\frac{1}{\tau_{\rm TG}} = \frac{1}{\tau_{\rm rec}} + D_a q^2. \tag{15}$$

In $1/\tau_{\rm TG} = 1/\tau_{\rm rec} + D_a q^2$, the intercept is $1/\tau_{\rm rec}$ and the slope is D_a.

The measured signals contain both a rapid initial decrease and a slower decay. Following the measurement instructions, we describe them by two exponentials and a constant background,

$$S(t,q) = C + A_f e^{-t/\tau_f} + A_s e^{-t/\tau_s}, \qquad 0 < \tau_f < \tau_s. \tag{16}$$

We use the slow rate $1/\tau_s$ to examine the dependence on q^2. The diffusion equation predicts a straight line if this component follows the same carrier population at every grating period. It does not, however, derive the two-exponential form. The two fitted times initially describe the shape of the measured curves; their physical interpretation must be checked against the results.

2.2 Ambipolar transport

For ambipolar transport, set the excess electron and hole densities equal, $n = h$, and equate their particle fluxes. With the drift–diffusion expressions

$$\mathbf{j}_e = -D_e \nabla n - \mu_e n \mathbf{E}, \qquad \mathbf{j}_h = -D_h \nabla n + \mu_h n \mathbf{E}, \tag{17}$$

the condition $\mathbf{j}_e = \mathbf{j}_h$ gives

$$-(D_e - D_h)\nabla n = (\mu_e + \mu_h) n \mathbf{E}, \tag{18}$$

$$\mathbf{E} = \frac{D_h - D_e}{\mu_e + \mu_h} \frac{\nabla n}{n}, \tag{19}$$

$$\mathbf{j}_e = -\left[D_e + \mu_e \frac{D_h - D_e}{\mu_e + \mu_h} \right] \nabla n \tag{20}$$

$$= -\frac{D_e \mu_h + D_h \mu_e}{\mu_e + \mu_h} \nabla n. \tag{21}$$

Comparing with Fick's law $\mathbf{j} = -D_a \nabla n$ identifies

$$D_a = \frac{D_e \mu_h + D_h \mu_e}{\mu_e + \mu_h}, \tag{22}$$

$$D_i = \frac{k_B T}{e} \mu_i \quad (i = e, h) \tag{23}$$

$$\implies \quad D_a = \frac{2\mu_e \mu_h}{\mu_e + \mu_h} \frac{k_B T}{e} = \mu_a \frac{k_B T}{e}, \tag{24}$$

$$\mu_a = \frac{2\mu_e \mu_h}{\mu_e + \mu_h}. \tag{25}$$

The Einstein step assumes non-degenerate carriers at a common temperature. At high carrier density, many-body and degeneracy effects can change D_a [7].

2.3 Pulse duration and spectral bandwidth

An autocorrelator splits each incoming laser pulse into two *pulse replicas*: copies that travel through separate interferometer arms and are then recombined. Their *optical delay* τ is the difference between their arrival times at the nonlinear crystal. For total propagation distances L_1 and L_2 in air, neglecting dispersion, $\tau \simeq (L_2 - L_1)/c$. Equal travel times give $\tau = 0$ and maximum temporal overlap. Moving one Michelson mirror by Δz changes the round-trip path length by $2\Delta z$, so the delay changes by $\Delta\tau = 2\Delta z/c$ (with the sign set by the chosen arm). Scanning this delay measures how the overlap changes, from which the pulse duration can be inferred under an assumed pulse shape.

For a Gaussian pulse intensity $I(t) = I_0 e^{-t^2/(2\sigma_t^2)}$, the intensity autocorrelation is

$$G(\tau) = \int_{-\infty}^{\infty} I(t) I(t - \tau)\, dt \tag{26}$$

$$= I_0^2 \int_{-\infty}^{\infty} \exp\left[-\frac{t^2 + (t - \tau)^2}{2\sigma_t^2} \right] dt \tag{27}$$

$$= I_0^2 e^{-\tau^2/(4\sigma_t^2)} \int_{-\infty}^{\infty} e^{-(t - \tau/2)^2/\sigma_t^2}\, dt \tag{28}$$

$$= I_0^2 \sqrt{\pi} \sigma_t e^{-\tau^2/(4\sigma_t^2)}. \tag{29}$$

Here $t^2 + (t - \tau)^2 = 2(t - \tau/2)^2 + \tau^2/2$ and the substitution $u = (t - \tau/2)/\sigma_t$ reduces the integral to $\sigma_t \int_{-\infty}^{\infty} e^{-u^2} du = \sqrt{\pi}\sigma_t$. The intensity half-maximum satisfies $e^{-t^2/(2\sigma_t^2)} = 1/2$, so $\Delta t_{\text{pulse}} = 2\sqrt{2 \ln 2}\, \sigma_t$. The autocorrelation half-maximum satisfies $e^{-\tau^2/(4\sigma_t^2)} = 1/2$, so $\Delta t_{\text{AC}} = 4\sqrt{\ln 2}\, \sigma_t = \sqrt{2}\, \Delta t_{\text{pulse}}$.

The Gaussian time–bandwidth product follows from the field envelope $a(t) = \sqrt{I_0} e^{-t^2/(4\sigma_t^2)}$. Set $F(\nu) = \int_{-\infty}^{\infty} a(t) e^{-i2\pi\nu t}\, dt$ and use $a'(t) = -t\, a(t)/(2\sigma_t^2)$. Differentiating and integrating by

parts on the real axis gives

$$F'(\nu) = -i2\pi \int_{-\infty}^{\infty} t\, a(t) e^{-i2\pi\nu t}\, dt \tag{30}$$

$$= i4\pi\sigma_t^2 \int_{-\infty}^{\infty} a'(t) e^{-i2\pi\nu t}\, dt \tag{31}$$

$$= i4\pi\sigma_t^2 \left([a(t)e^{-i2\pi\nu t}]_{-\infty}^{\infty} + i2\pi\nu F(\nu) \right) \tag{32}$$

$$= -8\pi^2\sigma_t^2 \nu F(\nu), \tag{33}$$

$$\int_{F(0)}^{F(\nu)} \frac{dF}{F} = -8\pi^2\sigma_t^2 \int_0^{\nu} \nu'\, d\nu', \tag{34}$$

$$F(\nu) = F(0) e^{-4\pi^2\sigma_t^2\nu^2}, \qquad F(0) = 2\sqrt{\pi}\sigma_t\sqrt{I_0}, \tag{35}$$

$$|F(\nu)|^2 = 4\pi\sigma_t^2 I_0 e^{-8\pi^2\sigma_t^2\nu^2}. \tag{36}$$

The boundary term vanishes because the Gaussian tends to zero at both limits; $F(0)$ follows by substituting $t = 2\sigma_t u$ in the Gaussian integral. At half maximum, $|F(\nu)|^2 = |F(0)|^2/2$, so

$$8\pi^2\sigma_t^2\nu^2 = \ln 2, \tag{37}$$

$$\Delta\nu = 2|\nu| = \frac{\sqrt{\ln 2}}{\sqrt{2}\pi\sigma_t}. \tag{38}$$

Combining this bandwidth with $\Delta t_{\text{pulse}} = 2\sqrt{2\ln 2}\,\sigma_t$ gives

$$\Delta t_{\text{pulse}} \Delta\nu = 2\sqrt{2\ln 2}\,\sigma_t \frac{\sqrt{\ln 2}}{\sqrt{2}\pi\sigma_t} \tag{39}$$

$$= \frac{2\ln 2}{\pi} \simeq 0.441. \tag{40}$$

From $\nu = c/\lambda$, differentiation gives $d\nu/d\lambda = -c/\lambda^2$; for a narrow spectrum centered at λ_0, $\Delta\nu \simeq (c/\lambda_0^2)\Delta\lambda$.

These relations allow us to compare the duration measured by autocorrelation with the shortest Gaussian pulse supported by the spectrum. A larger time–bandwidth product can arise from chirp, although errors in the pulse model or delay conversion can also contribute. The comparison assumes that the fitted spectrum is representative of the pulse and approximately Gaussian. The interferometric signal measured in this experiment contains additional oscillating terms, which we derive below.

2.4 Derivation of the measured interferometric autocorrelation

In interferometric autocorrelation (IAC), we measure the integrated second-harmonic signal while changing the delay between two pulse replicas. To obtain a pulse duration from this signal, we first derive the Gaussian overlap terms and then relate the optical delay to the recorded time. We use t for time within a pulse, τ for the optical delay between replicas, and x for the much slower oscilloscope time.

The oscilloscope records the signal as the mirror moves. The recorded-time separation $\Delta x = x_{i+1} - x_i$ is the time elapsed between two neighboring fringe maxima, expressed here in milliseconds. The optical delay between the pulses is instead on the femtosecond scale. Mirror motion connects these two times: if the optical path changes more quickly, more fringes pass in the same recorded interval. The laser's optical frequency is unchanged.

2.4.1 Measured signal and expansion of the tutorial formula

Assume identical replicas, an instantaneous second-order response, weak conversion, and phase matching over the relevant bandwidth. The second-harmonic field is proportional to the square of the combined fundamental field. A detector integrating over the pulse therefore records

$$S_{\mathrm{SHG}}(\tau) = \eta \int_{-\infty}^{\infty} \left|[E(t) + E(t-\tau)]^2\right|^2 dt = \eta \int_{-\infty}^{\infty} |E(t) + E(t-\tau)|^4 dt, \tag{41}$$

where η is a constant conversion and detection factor. This is tutorial Eq. (4.1). For comparison, its printed expansion, Eq. (4.2), is

$$\begin{aligned} I_{\mathrm{AC}}(\tau) = & \int_{-\infty}^{\infty} [I^2(t) + I^2(t-\tau)]\, dt \\ & + \int_{-\infty}^{\infty} [I(t) + I(t-\tau)]\, \mathrm{Re}\{E(t)E^*(t-\tau)\}\, dt \\ & + \int_{-\infty}^{\infty} \mathrm{Re}\{E^2(t)E^{*2}(t-\tau)\}\, dt \\ & + \int_{-\infty}^{\infty} I(t)I(t-\tau)\, dt. \end{aligned} \tag{4.2}$$

Here $I(t) = |E(t)|^2$. We denote the four integrals by T_0, T_1, T_2, T_3. To find their relative coefficients, we expand the fourth power explicitly. With

$$a = |E(t)|^2, \qquad b = |E(t-\tau)|^2, \qquad z = E(t)E^*(t-\tau), \qquad zz^* = ab. \tag{42}$$

Then

$$|E(t) + E(t-\tau)|^4 = (a + b + z + z^*)^2 \tag{43}$$

$$= a^2 + b^2 + 2ab + 2(a+b)(z+z^*) + z^2 + z^{*2} + 2zz^* \tag{44}$$

$$= a^2 + b^2 + 4ab + 4(a+b)\,\mathrm{Re}\, z + 2\,\mathrm{Re}(z^2). \tag{45}$$

The identities $z + z^* = 2\,\mathrm{Re}\, z$ and $z^2 + z^{*2} = 2\,\mathrm{Re}(z^2)$ follow by adding conjugate pairs; no real-field assumption is made. Integration gives

$$S_{\mathrm{SHG}} = \eta(T_0 + 4T_1 + 2T_2 + 4T_3). \tag{46}$$

The expansion gives relative factors of 1, 4, 2, and 4. These factors are missing from the printed Eq. (4.2), so we include them when calculating the signal. They cannot be absorbed into one common normalization.

2.4.2 Evaluation of the Gaussian overlap integrals

Take an unchirped Gaussian field with intensity standard deviation σ_t:

$$E(t) = E_0 e^{-t^2/(4\sigma_t^2)} e^{-i\omega_0 t}, \qquad I(t) = I_0 e^{-t^2/(2\sigma_t^2)}, \qquad I_0 = |E_0|^2. \tag{47}$$

The common Gaussian integral can be evaluated explicitly. Since its integrand is positive,

$$J = \int_{-\infty}^{\infty} e^{-v^2}\, dv > 0, \tag{48}$$

$$J^2 = \int_{\mathbb{R}^2} e^{-(v^2+w^2)}\, dv\, dw = \int_0^{2\pi} d\theta \int_0^{\infty} e^{-\rho^2} \rho\, d\rho \tag{49}$$

$$= 2\pi \left[-\tfrac{1}{2} e^{-\rho^2}\right]_0^{\infty} = \pi, \qquad J = \sqrt{\pi}. \tag{50}$$

For any real shift d, substituting $v = (t-d)/\sigma_t$, so $dt = \sigma_t\, dv$, gives

$$\int_{-\infty}^{\infty} e^{-(t-d)^2/\sigma_t^2} dt = \sigma_t \int_{-\infty}^{\infty} e^{-v^2} dv = \sqrt{\pi}\sigma_t. \tag{51}$$

Define $Q = I_0^2\sqrt{\pi}\sigma_t$. Translation invariance gives the constant term:

$$T_0 = I_0^2 \int_{-\infty}^{\infty} \left[e^{-t^2/\sigma_t^2} + e^{-(t-\tau)^2/\sigma_t^2}\right] dt = 2Q. \tag{52}$$

For the fundamental-frequency term, the carrier phases multiply as

$$z = E(t)E^*(t-\tau) = I_0 e^{-[t^2+(t-\tau)^2]/(4\sigma_t^2)} e^{-i\omega_0 t} e^{i\omega_0(t-\tau)} = I_0 e^{-[t^2+(t-\tau)^2]/(4\sigma_t^2)} e^{-i\omega_0\tau}. \tag{53}$$

In the definition of T_1, the real part is taken before integration. It can be moved outside the integral because $w(t) = I(t) + I(t-\tau)$ is real and the Gaussian integrals converge absolutely. At fixed τ,

$$\begin{aligned} T_1 &= \int_{-\infty}^{\infty} w(t)\,\mathrm{Re}\, z(t)\, dt \\ &= \frac{1}{2}\int_{-\infty}^{\infty} w(t)[z(t) + z^*(t)]\, dt \\ &= \frac{1}{2}\left[\int_{-\infty}^{\infty} w(t)z(t)\, dt + \left(\int_{-\infty}^{\infty} w(t)z(t)\, dt\right)^*\right] \\ &= \mathrm{Re}\int_{-\infty}^{\infty} w(t)z(t)\, dt. \end{aligned}$$

The third line uses $\int wz^*\, dt = (\int wz\, dt)^*$ for real w. Thus the complex overlaps may be integrated first and their real parts taken afterward, without assuming that z is real.

Completing the square separately for its two intensity factors gives

$$3t^2 + (t-\tau)^2 = 4(t-\tau/4)^2 + 3\tau^2/4, \tag{54}$$

$$t^2 + 3(t-\tau)^2 = 4(t-3\tau/4)^2 + 3\tau^2/4. \tag{55}$$

Consequently,

$$\int_{-\infty}^{\infty} I(t)z\, dt = I_0^2 e^{-i\omega_0\tau} e^{-3\tau^2/(16\sigma_t^2)} \int_{-\infty}^{\infty} e^{-(t-\tau/4)^2/\sigma_t^2} dt \tag{56}$$

$$= Qe^{-3\tau^2/(16\sigma_t^2)} e^{-i\omega_0\tau}, \tag{57}$$

$$\int_{-\infty}^{\infty} I(t-\tau)z\, dt = I_0^2 e^{-i\omega_0\tau} e^{-3\tau^2/(16\sigma_t^2)} \int_{-\infty}^{\infty} e^{-(t-3\tau/4)^2/\sigma_t^2} dt \tag{58}$$

$$= Qe^{-3\tau^2/(16\sigma_t^2)} e^{-i\omega_0\tau}. \tag{59}$$

Taking the real part of their sum, using $\mathrm{Re}\, e^{-i\theta} = \cos\theta$, yields

$$T_1 = 2Qe^{-3\tau^2/(16\sigma_t^2)} \cos(\omega_0\tau). \tag{60}$$

For the remaining two terms, use

$$t^2 + (t-\tau)^2 = 2(t-\tau/2)^2 + \tau^2/2. \tag{61}$$

The intensity overlap becomes

$$T_3 = I_0^2 \int_{-\infty}^{\infty} e^{-[t^2+(t-\tau)^2]/(2\sigma_t^2)} dt \tag{62}$$

$$= I_0^2 e^{-\tau^2/(4\sigma_t^2)} \int_{-\infty}^{\infty} e^{-(t-\tau/2)^2/\sigma_t^2} dt = Qe^{-\tau^2/(4\sigma_t^2)}. \tag{63}$$

Since $z^2 = I(t)I(t-\tau)e^{-2i\omega_0\tau}$, the second-harmonic overlap is

$$T_2 = \mathrm{Re}\int_{-\infty}^{\infty} z^2 dt = \mathrm{Re}\{e^{-2i\omega_0\tau}T_3\} = Qe^{-\tau^2/(4\sigma_t^2)}\cos(2\omega_0\tau). \tag{64}$$

Inserting all four terms into Eq. (46) gives

$$\begin{aligned} S_{\mathrm{SHG}}(\tau) = \eta Q\big\{2 + 8e^{-3\tau^2/(16\sigma_t^2)}\cos(\omega_0\tau) \\ + e^{-\tau^2/(4\sigma_t^2)}[4 + 2\cos(2\omega_0\tau)]\big\}. \end{aligned} \tag{65}$$

The cosine terms follow directly from the relative carrier phase of the two replicas. At zero delay, the result is $16\eta Q$, while for separated pulses it tends to $2\eta Q$. This gives the expected ideal interferometric contrast of $8:1$.

2.4.3 Pulse FWHM versus autocorrelation-envelope width

The pulse duration quoted in this report is the *intensity* FWHM. Its half-maximum condition gives

$$I(t_{1/2})/I_0 = e^{-t_{1/2}^2/(2\sigma_t^2)} = \tfrac{1}{2}, \tag{66}$$

$$t_{1/2}^2 = 2\sigma_t^2 \ln 2, \tag{67}$$

$$\Delta t_{\mathrm{pulse}} = 2|t_{1/2}| = 2\sqrt{2\ln 2}\,\sigma_t. \tag{68}$$

For the two distinct Gaussian overlap factors in Eq. (65),

$$g(\tau) = e^{-\tau^2/(4\sigma_t^2)}, \qquad h(\tau) = e^{-3\tau^2/(16\sigma_t^2)}, \tag{69}$$

$$g(\tau_{1/2}) = \tfrac{1}{2} \Rightarrow |\tau_{1/2}| = 2\sigma_t\sqrt{\ln 2}, \qquad \Delta\tau_g = 4\sigma_t\sqrt{\ln 2} = \sqrt{2}\,\Delta t_{\mathrm{pulse}}, \tag{70}$$

$$h(\tau_{1/2}) = \tfrac{1}{2} \Rightarrow |\tau_{1/2}| = 4\sigma_t\sqrt{\ln 2/3}, \qquad \Delta\tau_h = 8\sigma_t\sqrt{\ln 2/3} = \sqrt{8/3}\,\Delta t_{\mathrm{pulse}}. \tag{71}$$

The two overlap functions therefore have different widths. The factor $\sqrt{2}$ relates the isolated intensity autocorrelation $T_3 = Qg$ to the pulse duration. Since the full interferometric signal contains both g and h, applying this factor to the width of its complete envelope would give an incorrect duration.

The tutorial's alternative field notation $E(t) \propto e^{-t^2/\Gamma^2}$ corresponds to $\Gamma = 2\sigma_t$. Hence

$$\Delta t_{\mathrm{pulse}} = \Gamma\sqrt{2\ln 2}, \qquad \Delta t_{\mathrm{field}} = 2\Gamma\sqrt{\ln 2}. \tag{72}$$

Thus Γ itself is neither of these FWHM values. The spectral-intensity bandwidth and its time–bandwidth product are treated separately in the preceding subsection.

2.4.4 Recorded-time calibration and the fitted signal

The recorded time x is expressed in milliseconds relative to an initial peak estimate. Let x_0 be the fitted center and $r = x - x_0$. The optical-delay coordinate u, in carrier cycles, is modeled locally by

$$u(r) = fr + b_2r^2 + b_3r^3, \qquad \tau(x) = \frac{\lambda_0}{c}u(r), \tag{73}$$

$$\frac{du}{dx} = f + 2b_2r + 3b_3r^2, \qquad \omega_0\tau(x) = 2\pi u(r). \tag{74}$$

Here f, b_2, and b_3 have units of cycles/ms, cycles/ms^2, and cycles/ms^3, respectively. One fundamental fringe cycle corresponds to a delay change λ_0/c. A constant scan speed would give

$b_2 = b_3 = 0$; the changing fringe spacing in Fig. 13 motivates allowing a nonlinear mapping. Successive maxima initialize it, and the fitted derivative is checked to be positive throughout the sampled fit window. This is an empirical calibration rather than an independent measurement of mirror displacement.

The blue points in Fig. 13 are calculated from pairs of successive detected maxima in the raw signal. For maxima at recorded times x_i and x_{i+1}, their horizontal coordinate and vertical value are

$$x_{\mathrm{mid},i} = \frac{x_i + x_{i+1}}{2}, \qquad f_i = \frac{1}{x_{i+1} - x_i}. \tag{75}$$

Thus each point represents the inverse fringe spacing, in cycles per millisecond, placed halfway between its two peaks. Closer maxima give a higher value. The code uses SciPy's `find_peaks` function within the fitted window, with a prominence threshold of 5% of that window's raw-signal range and a minimum separation of approximately $0.6/f$, converted to sample indices. Here f is the fitted central fringe rate. The peak positions come from the measured samples, but their selection is therefore not fully independent of the fit. Noise, missed maxima, or extra detected maxima can affect these estimates; pointwise uncertainties have not been quantified. The red curve instead shows du/dx from the fitted cubic mapping. Agreement supports the delay calibration, but does not independently establish a particular mechanical trajectory.

Write the pulse-intensity standard deviation in optical cycles as $s = c\sigma_t/\lambda_0$. The overlap functions then become

$$g(u) = e^{-u^2/(4s^2)}, \qquad h(u) = e^{-3u^2/(16s^2)}. \tag{76}$$

The implemented fit uses every raw sample within 2.8 ms of the initial peak estimate:

$$\begin{aligned} S_{\mathrm{fit}}(x) = B + A_0 g(u) + h(u)[C_1 \cos(2\pi u) + D_1 \sin(2\pi u)] \\ + g(u)[C_2 \cos(4\pi u) + D_2 \sin(4\pi u)]. \end{aligned} \tag{77}$$

The baseline and harmonic coefficients are independent fit parameters. In the ideal model, with matching signal units and phase origin, they would satisfy

$$B = 2\eta Q, \quad A_0 = 4\eta Q, \quad C_1 = 8\eta Q, \quad C_2 = 2\eta Q, \quad D_1 = D_2 = 0. \tag{78}$$

In the fit, we allow these coefficients to vary so that the measured amplitudes and phases need not match the ideal values. The Gaussian overlap widths are retained. This makes the fit more flexible, but does not determine the detector response or account fully for chirp.

The fitted width is converted directly to the pulse duration:

$$\sigma_t = s\lambda_0/c, \qquad \boxed{\Delta t_{\mathrm{pulse}} = 2\sqrt{2\ln 2}\, s\lambda_0/c.} \tag{79}$$

We thus obtain the pulse duration directly from the fitted s. The conversion does not require measuring an envelope width. The associated statistical errors and the remaining calibration and model uncertainties are discussed in Sec. 3.4.

3 Experimental setup and measurement

3.1 Optical arrangement

The optical arrangement is shown in Fig. 2. According to the tutorial, the pulses are produced by a Kerr-lens mode-locked Ti:sapphire laser. Self-focusing favors pulsed operation, and a prism compressor compensates dispersion accumulated along the optical path. To measure the pulse duration, a Michelson interferometer splits the pulse into two replicas. Their delay is varied by moving one mirror, and their overlap is detected through second-harmonic generation in BBO, as shown in Fig. 3.

For the carrier measurements, the light is divided into pump and probe paths. The two pump beams cross on the GaAs sample and produce the excitation grating. In the BOXCAR arrangement, the diffracted probe interferes with a local oscillator, giving a signal linear in the weak diffracted field. Figure 4 shows this detection principle and its phase dependence. A rotating coverslip varies the local-oscillator phase, while a chopper modulates the pump. The lock-in amplifier extracts the response at the chopper frequency.

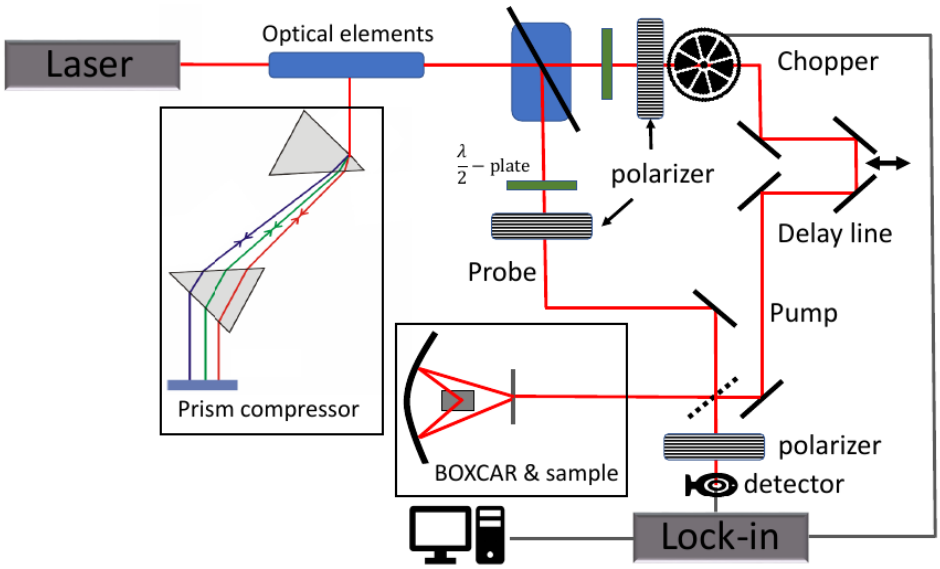


Figure 2: Tutorial schematic of the transient-reflectivity and transient-grating apparatus, including prism compressor, pump and probe paths, delay line, chopper, BOXCAR sample geometry, detector, and lock-in amplifier. Reproduced from Fig. 4.7 of the course tutorial.

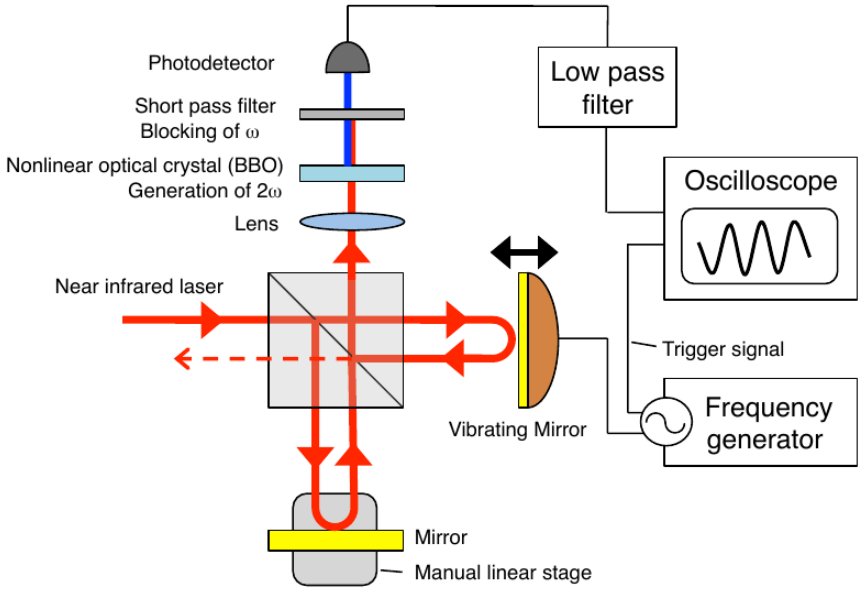


Figure 3: Michelson autocorrelator schematic: the vibrating mirror varies the relative delay between pulse replicas, and a BBO crystal generates the detected second harmonic. Reproduced from Fig. 4.1 of the course tutorial.

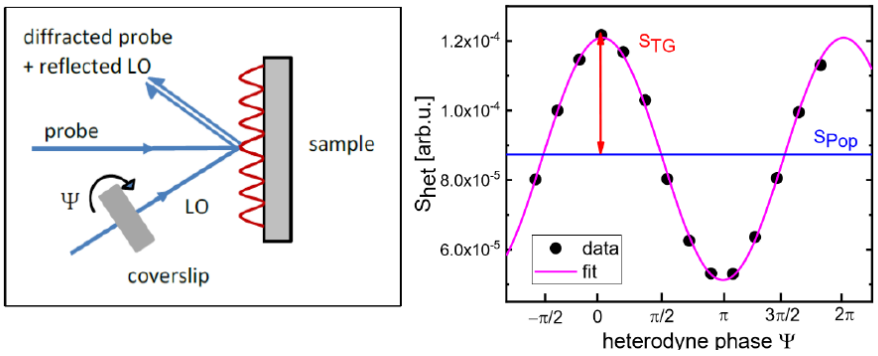


Figure 4: BOXCAR heterodyne detection of the diffracted probe and reflected local oscillator, with the measured signal illustrated as a function of heterodyne phase. Reproduced from Fig. 4.6 of the course tutorial.

3.2 Measurement data and calibration

The measurements used here consist of the spectrum in `spektrum.csv`, three autocorrelation traces before compression (`vor1`–`vor3`), and three afterwards (`nach1`–`nach3`). There is also one PP dataset and one TG dataset for each of four grating periods. In these transient files, the first column contains the delay in picoseconds and the remaining columns contain repeated scans.

We take the supplied PP/TG delays and grating spacings as accurate and use them as exact inputs. The numbers 5.6, 6.5, 8, and 12 in the TG filenames give the phase-mask periods in micrometres. According to the accompanying email, the optical system reduces these periods by a factor of four at the sample. The sample period and wave number are therefore

$$\Lambda_{\mathrm{sample}} = \frac{\Lambda_{\mathrm{mask}}}{4}, \qquad q = \frac{2\pi}{\Lambda_{\mathrm{sample}}} = \frac{8\pi}{\Lambda_{\mathrm{mask}}}.$$

For example, the 12 µm mask gives a 3 µm grating on the sample and $q = 2.094 \times 10^{6}\,\mathrm{m}^{-1}$, rounded for display. Table 2 lists all four conversions. Since the periods and demagnification are taken as exact, no additional uncertainty is assigned to q and the incidence angles are not needed for this calculation.

The horizontal coordinates in Fig. 11 thus come from the grating geometry. To obtain the vertical coordinates, we average the retained scans, fit their decay, and calculate the slow rate as $k_s = 1/\tau_s$. The PP measurement has no imposed periodic grating and is placed at $q = 0$. Treating it as a uniform-population reference also assumes that diffusion out of the finite probe region can be neglected.

Figure 5 compares the autocorrelation records. The sampling interval and total duration of `vor2` are five times shorter than those of the other records. Its oscillations also remain visible at the end of the measurement, so the signal is not fully covered. We quantify this truncation in Part I. For the remaining traces, we compare the residual root-mean-square (RMS), divided by the signal range in the fitted interval, and display the best-fitting trace from each group. This selects the smallest relative residual, rather than proving which trace has the smallest physical uncertainty.

3.3 Data screening and fit procedure

Before averaging, we compare the repeated scans to identify measurements that differ strongly from the rest. For TG datasets with at least eight repeats, each scan is compared with the pointwise median. We divide its deviations by the larger of the robust scale 1.4826 MAD and the lower quartile of that scale, then calculate their RMS. Here MAD denotes the median absolute deviation. From the resulting scores s_j, we calculate $z_j = 0.6745(s_j - \mathrm{median}(s))/\mathrm{MAD}(s)$ and

flag scans with $z_j > 3.5$. The full flagged scans are excluded; individual delay points are retained. Figure 10 compares the excluded scans with the others. Since a statistical deviation does not prove an acquisition fault, we also repeat the diffusion analysis with these scans included.

We average the retained scans at each delay and calculate the standard error of the mean. The averaged curves are then fitted with the two-exponential function in Sec. 2.1, from 0.5 ps to the final recorded delay. We use unweighted least squares, so every recorded point has equal weight even where the delay spacing changes. The error bars show the repeat-scan standard errors. The PP record extends to 2000 ps, while the TG records end between 26.6 ps and 57 ps; we therefore also examine the effect of changing the fitting interval.

To estimate the uncertainty from repeat measurements, we use a bootstrap with 300 resamples (Appendix D.1). Each resample selects complete scans and repeats the averaging, fitting, and diffusion regression. The resulting 95% repeat-scan intervals are explained in Appendix D.2. They describe repeat variability for the chosen model and scan selection, without including uncertainty in the model itself.

The files do not contain measured sample temperature, pump and probe fluence, spot size, sample thickness and orientation, incidence angles, or detector and lock-in settings. The tutorial and the `RT` filenames indicate room-temperature measurements, but no temperature reading is available. This missing information limits the comparison with literature. The Python analysis and numerical results are stored in `analysis/`.

3.4 Uncertainty conventions and propagation

We assess the fits using their residuals and, where repeat errors are available, the reduced chi-squared χ^2_ν. Appendix D.4 gives the calculation and explains its limitations for these unweighted fits. For the spectrum and IAC, pointwise measurement errors are unknown, so a reduced chi-squared cannot be calculated independently.

Throughout the report, $\pm$ denotes one standard error unless stated otherwise. For nonlinear fits, it is calculated from the parameter covariance matrix; for straight-line fits, it is the ordinary least-squares regression error. The bracketed 95% intervals for transient lifetimes and diffusivity instead come from the repeat-scan bootstrap. The two measures describe different sources of variation. Neither includes uncertainty in the physical interpretation, trace selection, or temperature. For the pulse measurements, wavelength calibration and the fitted delay conversion introduce further uncertainty that has not been quantified. The supplied PP/TG delays and grating spacings are treated as exact.

For a derived quantity $y = f(\mathbf{p})$ with covariance matrix Σ, first-order propagation uses

$$u_y^2 = \nabla f^{\mathsf{T}} \Sigma \nabla f. \tag{80}$$

The full spectral covariance is retained for the peak level $A + B$, half-maximum level $A/2 + B$, frequency bandwidth, and transform-limit duration. If s is the fitted pulse width in carrier cycles, then $\Delta t \propto s\lambda_0$ and $\Delta\nu \propto \sigma_\lambda/\lambda_0^2$, so the time–bandwidth product is proportional to $s\sigma_\lambda/\lambda_0$. Its error propagation therefore includes the shared wavelength dependence rather than treating the derived duration and bandwidth as independent. Spectrum and IAC fit parameters from separate records are treated as independent. The common wavelength cancels in the before/after duration ratio. For the nominal intercept lifetime, $u_{1/b} = u_b/b^2$ is a first-order standard error, not a symmetric 95% interval.

Theoretical constants, counts, file labels, selection thresholds, and chosen fitting intervals are specified inputs. We also identify descriptive quantities such as RMS and scan scores when their uncertainties have not been estimated. Treating the supplied transient delays as exact does not determine the autocorrelation delay: the oscilloscope timestamps must still be converted using the fringes, as derived in Sec. 2.4. The error propagation is implemented in `analysis/report_uncertainties.py`.

4 Analysis and results

4.1 Part I: Pulse duration and spectral bandwidth

We first compare the autocorrelation traces before and after compression in Fig. 5. The `vor2` signal is still oscillating strongly at the right boundary. To measure this, we calculate the RMS deviation from the whole-record median over the last 10% of each record. The values are 0.397 recorded units for `vor2`, compared with 0.0081 and 0.0058 for `vor1` and `vor3`. Their uncertainties have not been estimated, and the median serves only as a reference level. This agrees with the visible truncation: `vor2` provides less information about the wings and background. We therefore exclude it from the pulse-duration calculation and use the more complete records.

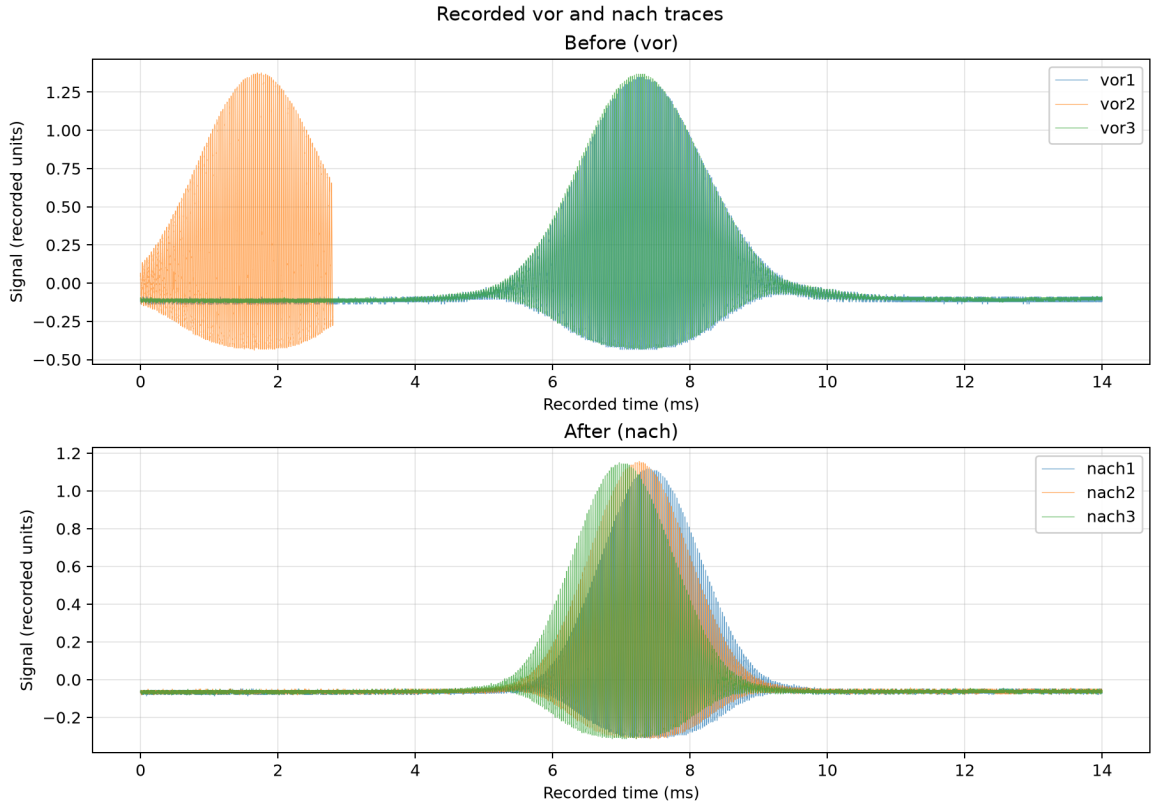


Figure 5: Recorded auxiliary traces grouped as before (vor) and after (nach). The horizontal axis uses recorded time converted to milliseconds.

The measured spectrum and Gaussian fit are shown in Fig. 6. The nominal wavelength coordinates (calibration uncertainty unavailable) span approximately 345 nm to 1041 nm. We fitted all recorded points with

$$S(\lambda) = B + A \exp\left[-\frac{(\lambda - \lambda_0)^2}{2\sigma_\lambda^2}\right]. \tag{81}$$

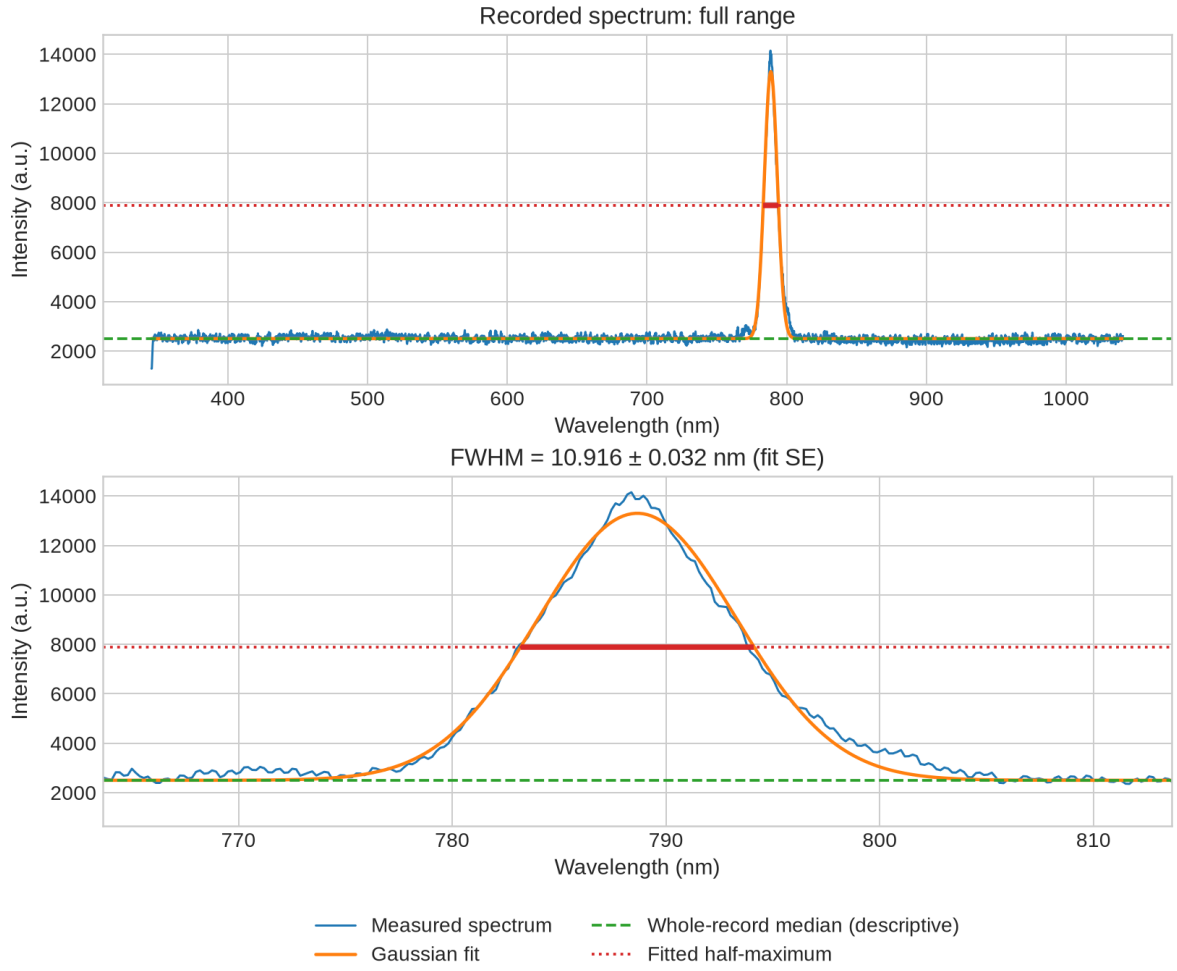
The peak above baseline is A and the fitted peak level is $B + A$. At half peak height above baseline, $S - B = A/2$, so

$$e^{-(\lambda-\lambda_0)^2/(2\sigma_\lambda^2)} = \frac{1}{2}, \tag{82}$$

$$|\lambda - \lambda_0| = \sqrt{2\ln 2}\,|\sigma_\lambda|, \tag{83}$$

$$\Delta\lambda_{\mathrm{FWHM}} = 2\sqrt{2\ln 2}\,|\sigma_\lambda|. \tag{84}$$

The fit gives a centre of (788.656 ± 0.013) nm, peak intensity of (13302 ± 27) a.u., and fitted half-maximum level of (7902 ± 14) a.u., with $\Delta\lambda_{\mathrm{FWHM}} = (10.916 \pm 0.032)$ nm. Propagating the full spectral covariance through $\Delta\nu \simeq c\Delta\lambda/\lambda_0^2$ gives $\Delta\nu = (5.261 \pm 0.015)$ THz and $\Delta t_{\mathrm{TL}} = (2\ln 2/\pi)/\Delta\nu = (83.87 \pm 0.24)$ fs. All quoted errors here are formal one-standard-error uncertainties, excluding calibration and model uncertainty. The median measured intensity is approximately 2501 a u ; its measurement uncertainty is unquantified because repeat spectra and an instrument-noise model are unavailable. The median is a descriptive statistic of all recorded wavelengths, not the fitted background B.



$$I(\lambda) = B + A\exp[-(\lambda - \lambda_0)^2/(2\sigma_\lambda^2)].$$

Quantity	Estimate	Quantity	Estimate
A (a.u.)	10802 ± 27	B (a.u.)	2500.7 ± 2.5
λ_0 (nm)	788.656 ± 0.013	σ_λ (nm)	4.636 ± 0.014
Peak (a.u.)	13302 ± 27	Half-max. (a.u.)	7902 ± 14
FWHM (nm)	10.916 ± 0.032		
χ^2_ν	unavailable		

χ^2_ν is unavailable without measurement-error estimates; see Appendix D.4.
Errors are conditional one-standard-error estimates; model uncertainty is excluded.

Figure 6: Measured spectrum, Gaussian fit, median level, and fitted half-maximum bandwidth. The displayed bandwidth error is a conditional fit standard error; raw-signal and wavelength-calibration uncertainties are unavailable.

We then fit the autocorrelation records with Eq. (77). The resulting pulse durations are

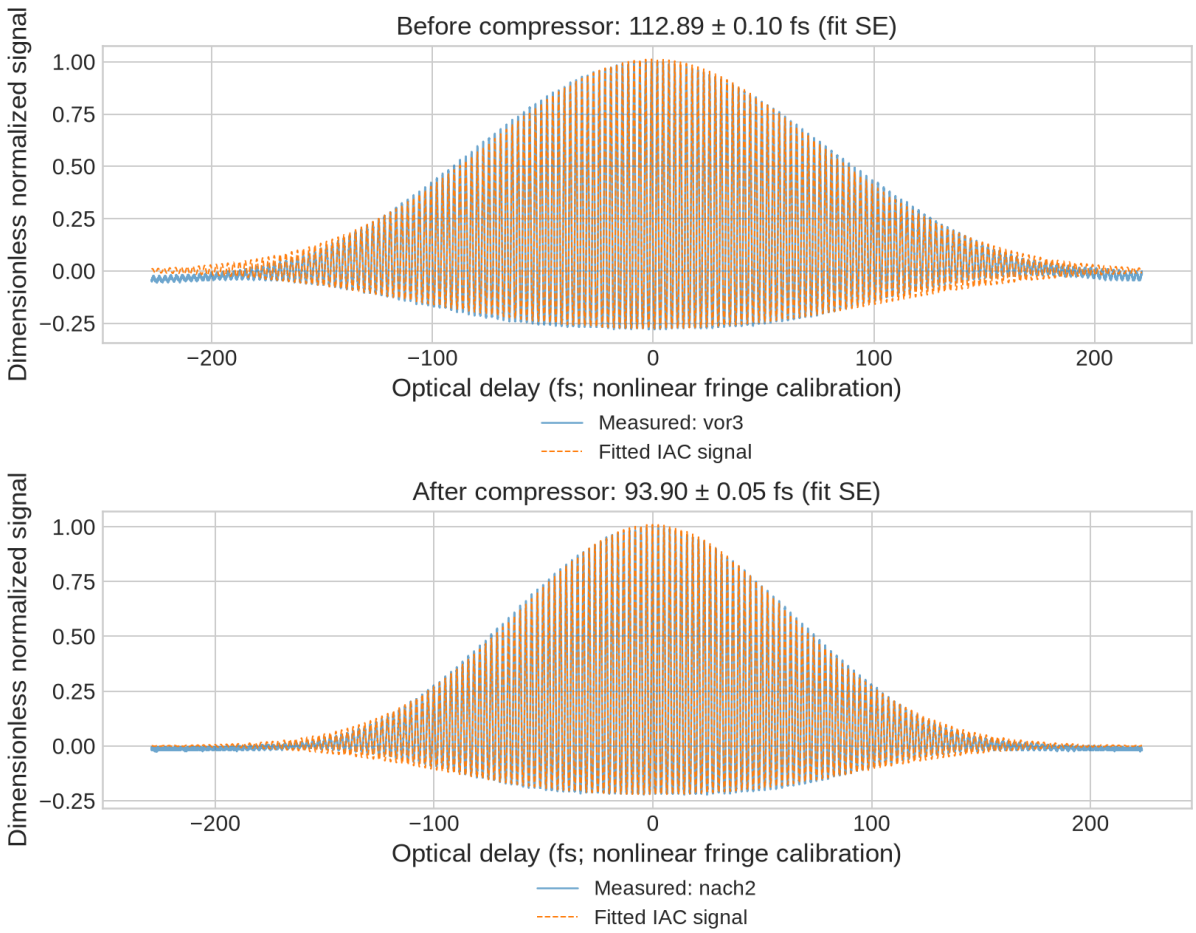
listed in Table 1. Of the traces before and after compression, `vor3` and `nach2` have the smallest normalized residuals and are shown in Fig. 7. Their durations are (112.89±0.10) fs and (93.902± 0.054) fs, respectively. The corresponding time–bandwidth products are 0.5940 ± 0.0018 and 0.4941 ± 0.0015, both above the Gaussian limit $2 \ln 2/\pi$. The duration decreases by (16.823 ± 0.089)%, consistent with the action of the compressor. These results refer to the selected traces, rather than averages of the repeats. The errors follow from the fit covariance and include the shared wavelength dependence; pulse-model and wavelength-calibration uncertainties remain unquantified.

Table 1: Gaussian pulse-intensity FWHM from the raw IAC fits. Errors are conditional covariance standard errors at fixed wavelength, excluding calibration and model uncertainty. Normalized RMS is residual RMS divided by the measured range; its uncertainty has not been estimated.

Trace	Pulse FWHM (fs)	Normalized RMS	χ^2_ν
vor1	113.09 ± 0.10	0.0230	unavailable
vor3 (selected)	112.89 ± 0.10	0.0223	unavailable
nach1	93.99 ± 0.05	0.0135	unavailable
nach2 (selected)	93.90 ± 0.05	0.0134	unavailable
nach3	93.95 ± 0.05	0.0135	unavailable

Figure 13 in Appendix A shows why the delay conversion matters. The measured fringe maxima become closer together near the center of the scan, so equal intervals of recorded time cannot be treated as equal optical delays. Fitting the same intervals with a constant scan speed gives durations of (74.53 ± 0.39) fs and (71.88 ± 0.28) fs, with conditional standard errors. The normalized residual RMS values are then 0.107 and 0.080, compared with 0.0223 and 0.0134 for the cubic delay conversion; uncertainties on these diagnostics have not been estimated. The constant-speed assumption therefore describes the measured fringes poorly and underestimates the duration. We choose the nonlinear conversion from the fringe behavior and residuals.

Varying the fit half-window from 2.0 ms to 3.2 ms changes the fitted duration from (111.65 ± 0.08) fs to (114.07 ± 0.10) fs for `vor3`, and from (92.69 ± 0.05) fs to (94.32 ± 0.05) fs for `nach2` (conditional fit standard errors at fixed wavelength). The differences quantify sensitivity to the tested window choices; they are not confidence intervals for the pulse duration. The shifts exceed the conditional covariance errors of 0.10 fs and 0.05 fs. The errors assume independent residuals and an exact model; structured residuals, detector filtering, optical chirp, wavelength calibration, and trace selection are not included. Likewise, restricting the spectral fit to windows of ±50, ±30, and ±20 nm about 789 nm changes the FWHM to (10.783±0.053), (10.651±0.065), and (10.434± 0.078) nm, respectively, compared with (10.916 ± 0.032) nm for the full record (conditional fit standard errors). This background/window dependence exceeds the formal 0.03 nm covariance error. These sensitivity ranges are diagnostics, not confidence intervals. The residuals are shown in Appendix A, Fig. 12.



$$
\begin{aligned}
S_{\mathrm{fit}} &= B + A_0 g + h[C_1 \cos(2\pi u) + D_1 \sin(2\pi u)] \\
&\quad + g[C_2 \cos(4\pi u) + D_2 \sin(4\pi u)], \\
g &= e^{-u^2/(4s^2)}, \quad h = e^{-3u^2/(16s^2)}, \\
u &= f(x - x_0) + b_2(x - x_0)^2 + b_3(x - x_0)^3.
\end{aligned}
$$

Parameter	vor3	nach2
B (recorded units)	-0.03824 ± 0.00036	-0.04493 ± 0.00016
A_0 (recorded units)	0.36526 ± 0.00069	0.34501 ± 0.00034
C_1 (recorded units)	-0.23 ± 0.14	0.663 ± 0.027
D_1 (recorded units)	-0.876 ± 0.035	-0.326 ± 0.055
C_2 (recorded units)	-0.131 ± 0.025	0.074 ± 0.017
D_2 (recorded units)	0.080 ± 0.041	-0.106 ± 0.012
x_0 (ms)	0.00869 ± 0.00072	0.00247 ± 0.00038
s (cycles)	18.224 ± 0.016	15.1583 ± 0.0087
f (cycles/ms)	34.05357 ± 0.00031	34.25480 ± 0.00023
b_2 (cycles/ms^2)	-0.1140 ± 0.0010	-0.12715 ± 0.00055
b_3 (cycles/ms^3)	-0.45726 ± 0.00016	-0.45029 ± 0.00019
Pulse FWHM (fs)	112.89 ± 0.10	93.902 ± 0.054
χ^2_ν	unavailable	unavailable

Coefficients refer to the unnormalized recorded signal; the central wavelength is held fixed for the pulse-width errors. χ^2_ν requires unavailable pointwise error estimates (Appendix D.4).

Errors are conditional one-standard-error estimates; model uncertainty is excluded.

Figure 7: Selected IAC measurements (blue) and fitted signals (orange dashed). Both use the fitted-baseline subtraction and the same measured maximum-deviation normalization. The fit model, parameters, and conditional standard errors are given above.

4.2 Part II: Hot-carrier cooling and electron–phonon coupling

The second part concerns the energy lost by photoexcited carriers to the lattice. Following the assignment in the measurement instructions, we examine the fast PP component as a possible cooling time. The fit described in Part III gives $\tau_{\mathrm{cool}}^{\mathrm{eff}} = \tau_f = 0.717\,\mathrm{ps}$, with a 95% repeat-scan bootstrap interval of $[0.698, 0.743]$ ps. For this time to describe cooling, the fast reflectivity change must follow the excess carrier temperature. No calibration establishing this relation is available. The fit begins at 0.5 ps, excluding the earliest thermalization but still covering picosecond cooling. The tutorial distinguishes roughly 100 fs carrier thermalization from electron–lattice equilibration over a few picoseconds; uncertainties on these estimates are not supplied [3].

The fitted time also depends on how the PP response is described. Adding a third, slow exponential changes the fast time to (0.674 ± 0.011) ps, with a covariance standard error (Sec. 4.4.1). This change and the systematic residuals show that repeat variability is only one source of uncertainty. We therefore use the fast component as an effective cooling-time estimate under the stated optical-response assumption. Its physical assignment is not included in the statistical interval.

To relate a cooling time to electron–phonon coupling, we assume constant volumetric heat capacities C_e and C_l for the electrons and lattice. With a coupling coefficient G, their energy exchange gives

$$C_e\dot{T}_e = -G(T_e - T_l), \qquad C_l\dot{T}_l = G(T_e - T_l), \tag{85}$$

$$\Delta T = T_e - T_l, \qquad \frac{d\Delta T}{dt} = -G\left(\frac{1}{C_e} + \frac{1}{C_l}\right)\Delta T, \tag{86}$$

$$\int_{\Delta T(0)}^{\Delta T(t)} \frac{d\Delta T'}{\Delta T'} = -G\left(\frac{1}{C_e} + \frac{1}{C_l}\right)\int_0^t dt', \tag{87}$$

$$\Delta T(t) = \Delta T(0)e^{-t/\tau_{\mathrm{cool}}}, \qquad G = \frac{C_eC_l}{C_e + C_l}\frac{1}{\tau_{\mathrm{cool}}}. \tag{88}$$

If $C_l \gg C_e$ and $\tau_{\mathrm{cool}} = \tau_f$, then $G/C_e \simeq 1/\tau_f = 1.394\,\mathrm{ps}^{-1}$, with a conditional bootstrap interval $[1.347, 1.432]$ ps^{-1}. This is a relaxation rate under explicit assumptions, not a measured volumetric coupling coefficient in $\mathrm{W\,m^{-3}\,K^{-1}}$. For nondegenerate electrons of fixed density n, $C_e = \partial(3nk_BT_e/2)/\partial T_e = 3nk_B/2$, but n and the relevant temperature dependence are unknown here. A numerical G is therefore not determined.

The assignment also depends on the fit window: fitting PP only through 33 ps gives $\tau_f = (0.579 \pm 0.010)$ ps, whereas starting the full-record fit at 1 ps gives (0.96 ± 0.13) ps (conditional fit standard errors). These changes exceed the repeat-scan interval. For context, Hunsche et al. report thermalization within 200 fs, while Kash et al. infer an average electron–LO-phonon scattering time near 165 fs below 10^{17} cm^{-3} [4, 6]. The quoted 200 fs is a reported upper bound and 165 fs is an approximate literature value; associated uncertainties are not supplied in the cited abstracts. Neither microscopic time is the same observable as the effective temperature-relaxation time or our reflectivity fit component.

4.3 Part III: Transient-reflectivity and transient-grating decay times

We apply the scan comparison to the 44 TG repeats: 12 for the mask labeled 5.6, ten for 6.5, 12 for 8, and ten for 12. Only scans 4 and 10 of the period-8 dataset exceed the threshold of 3.5. Their scores are 3.51 and 5.26, respectively, with unestimated uncertainties. Scan numbers count the signal columns from one, after the delay column. Excluding these two complete scans leaves ten repeats for this period. We retain all other TG scans and all five PP scans; the latter are below the minimum of eight repeats required by the screening rule.

Figure 10 shows the two excluded traces. Scan 4 lies only just above the threshold, whereas scan 10 differs more strongly from the retained population. This supports examining their effect,

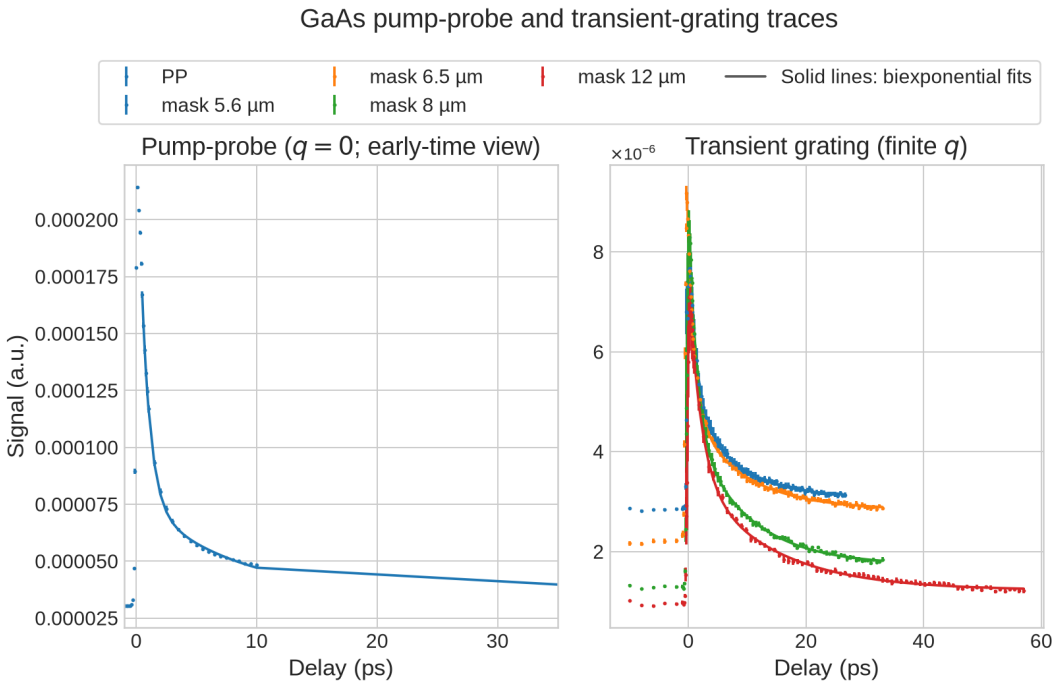
but does not identify the cause of the deviation. In Part IV we compare the diffusion result with and without these exclusions.

The averaged signals and their two-exponential fits are shown in Fig. 8. Figure 9 shows the residuals divided by the signal range of each fitting interval. Their RMS is 1.26% for PP and 0.71–0.81% for TG, with unestimated diagnostic uncertainties. Although these values are small compared with the full signal, the PP residuals show a systematic decrease at long delays. The signal continues to change after both fitted exponentials have decayed, while the fitted background remains constant. This behavior appears in every PP repeat, suggesting a limitation of the model rather than an isolated bad scan. We retain the PP measurement for comparison and fit the diffusion line to the finite-q TG data alone. Each residual panel uses its own delay range because the PP record is much longer than the TG records.

The fitted times are listed in Table 2, together with the 95% intervals obtained by resampling the retained scans. Both TG lifetimes tend to increase with grating period. The sample periods are calculated using the factor-of-four reduction described above, and all delays are in picoseconds. These coordinates are treated as exact; the PP measurement is represented by $q = 0$.

Table 2: Biexponential fit lifetimes with 95% repeat-scan bootstrap confidence intervals. Each lifetime is written as the fitted value followed by [lower bound, upper bound], in ps. The bounds are the 2.5th and 97.5th percentiles of fits obtained by resampling complete retained scans. They quantify repeat-scan variability conditional on the selected scans and model; model uncertainty is excluded. The filename value is the phase-mask period; the sample period is one quarter of that value. Supplied delay coordinates, periods, and derived wave numbers are treated as exact.

Data	$\Lambda_{\mathrm{sample}}$ (µm)	q ($10^6\,\mathrm{m}^{-1}$)	τ_f (ps)	τ_s (ps)
Pump–probe	–	0	0.717 [0.698, 0.743]	7.878 [7.753, 8.040]
TG 5.6	1.400	4.488	1.206 [1.031, 1.445]	7.205 [6.412, 8.771]
TG 6.5	1.625	3.867	1.298 [1.157, 1.503]	9.200 [8.156, 10.960]
TG 8	2.000	3.142	1.363 [1.064, 1.983]	9.496 [7.933, 12.250]
TG 12	3.000	2.094	2.060 [1.708, 2.463]	13.283 [11.970, 15.170]



$$S_{\mathrm{fit}}(t) = C + A_f e^{-t/\tau_f} + A_s e^{-t/\tau_s}.$$

Trace	τ_f (ps)	τ_s (ps)	A_f	A_s	C
PP	0.717 ± 0.042	7.88 ± 0.79	182.9 ± 6.5	42.3 ± 3.2	35.22 ± 0.28
TG 5.6	1.206 ± 0.056	7.20 ± 0.35	3.294 ± 0.069	2.042 ± 0.076	3.089 ± 0.015
TG 6.5	1.298 ± 0.063	9.20 ± 0.45	3.071 ± 0.062	1.917 ± 0.062	2.838 ± 0.016
TG 8	1.363 ± 0.067	9.50 ± 0.50	4.458 ± 0.093	2.793 ± 0.094	1.729 ± 0.025
TG 12	2.060 ± 0.095	13.28 ± 0.63	3.875 ± 0.085	2.381 ± 0.091	1.225 ± 0.016

Diagonal χ^2_ν diagnostics:

PP	TG 5.6	TG 6.5	TG 8	TG 12
120 ($\nu = 59$)	0.173 ($\nu = 65$)	0.292 ($\nu = 65$)	0.937 ($\nu = 65$)	0.662 ($\nu = 71$)

A_f, A_s, C are in 10^{-6} recorded signal units. These covariance standard errors differ from the 95% repeat-scan intervals in Table 2. χ^2_ν uses repeat-mean errors; diagnostic uncertainties are unestimated (Appendix D.4).

Errors are conditional one-standard-error estimates; model uncertainty is excluded.

Figure 8: Mean delay traces over retained scans and matching-color biexponential fits. The PP panel displays the first 35 ps for readability; its fit uses the full record through 2000 ps. Error bars show standard errors across retained scans.

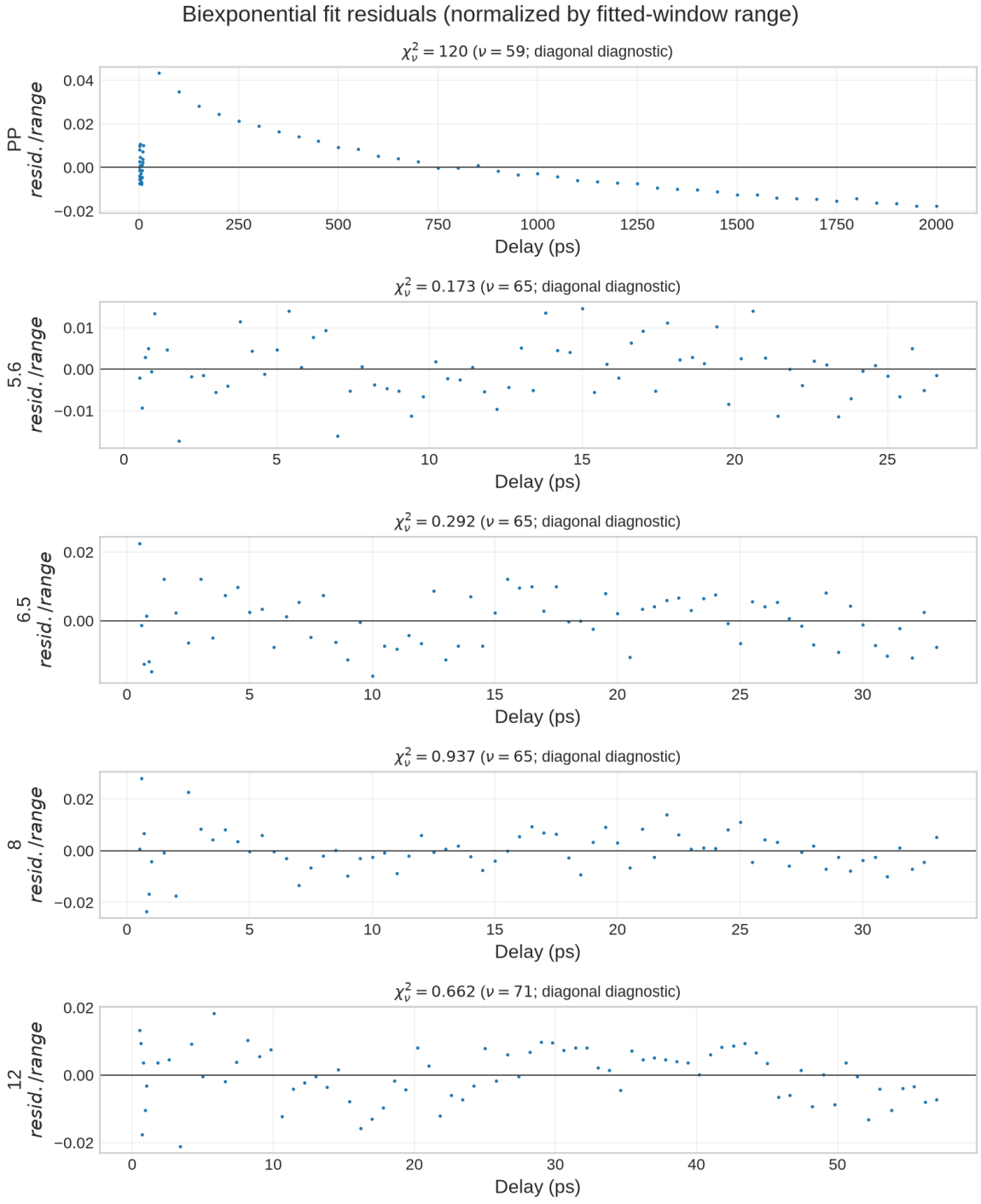


Figure 9: Biexponential fit residuals for the averaged data, normalized by the fitted-window signal range. Panels show diagonal χ^2_ν diagnostics and nominal degrees of freedom; their uncertainties are unestimated (Appendix D.4). Each panel has its own delay range.

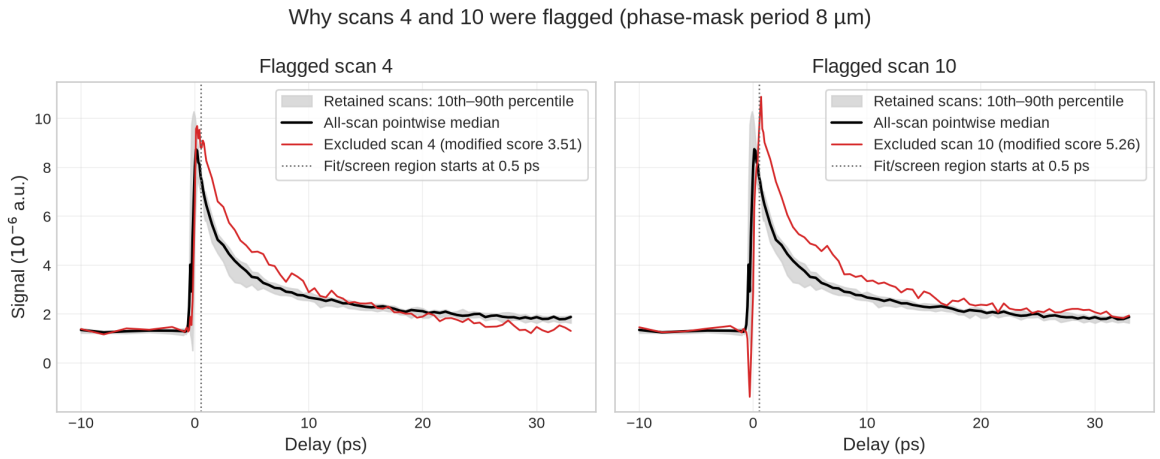


Figure 10: The two statistically flagged scans from the phase-mask-period-8 file, compared with the all-scan pointwise median and the 10th–90th percentile band of retained scans. The vertical dotted line marks the 0.5 ps fit/screen threshold. Scan 4 is borderline; scan 10 shows the larger deviation. Uncertainties on the displayed diagnostic scores have not been estimated.

The fitted lifetimes depend on the observation window. With the PP start held at 0.5 ps, its slow lifetime changes from (2.93 ± 0.11) ps when fitting through 33 ps, to (4.91 ± 0.29) ps through 57 ps, (5.33 ± 0.28) ps through 100 ps, and (7.88 ± 0.79) ps through 2000 ps (conditional covariance standard errors for this comparison). Holding the end at 2000 ps but moving the start to 2 ps gives (3.43 ± 0.19) ps and (375 ± 50) ps (conditional covariance standard errors); the two components then describe different portions of the relaxation. These covariance errors are distinct from the repeat-scan bootstrap intervals in Table 2 and do not account for systematic model mismatch. Consequently, the labels "fast" and "slow" do not identify invariant physical channels across arbitrary fit windows.

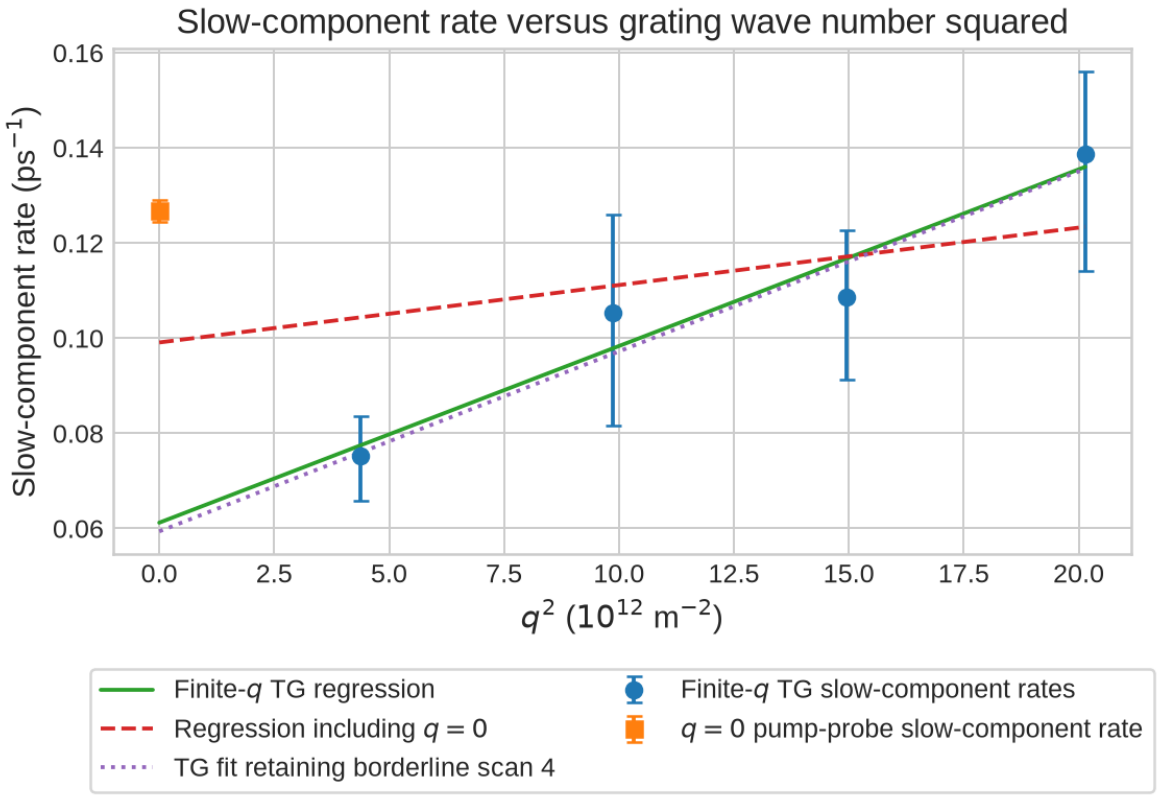
4.4 Part IV: Recombination, ambipolar diffusivity, and mobility

We now use the fitted slow TG lifetimes to examine diffusion. For each grating period, we calculate $k_s = 1/\tau_s$ and plot it against q^2 in Fig. 11. The diffusion model predicts a straight line. We describe its agreement with the data using the coefficient of determination R^2, explained in Appendix D.3.

For the finite-q TG measurements, the slow-component rates are regressed by unweighted least squares against q^2 as motivated by the diffusion equation. The slope is $(3.71 \pm 0.70) \times 10^{-15}\,\mathrm{m^2\,ps^{-1}}$ ($R^2 = 0.934$; diagnostic uncertainty not estimated), giving an effective ambipolar diffusivity inferred from the slow TG component, $D_a = (3.71 \pm 0.70) \times 10^{-3}\,\mathrm{m^2\,s^{-1}}$, where the stated error is the standard error of the four-point regression slope. Resampling complete retained scans gives a 95% bootstrap interval $[2.31, 4.96] \times 10^{-3}\,\mathrm{m^2\,s^{-1}}$. This interval includes repeat-scan variation but not model uncertainty. Retaining borderline scan 4 while still excluding scan 10 gives $(3.78 \pm 0.56) \times 10^{-3}\,\mathrm{m^2\,s^{-1}}$ (Fig. 11).

The slow TG rates follow the expected trend: a larger wave number gives a faster decay. However, identifying the slope with a material diffusion coefficient requires the slow component to represent the same population at every period. The PP comparison and the different fast- and slow-component slopes below do not confirm this assumption. We therefore refer to the result as the effective diffusivity of the slow TG component. Its model uncertainty is not included in the quoted errors.

Retaining both flagged scans also gives $D_a = (3.78 \pm 0.57) \times 10^{-3}\ \mathrm{m^2/s}$. The diffusion slope is therefore insensitive to these two exclusions at the quoted precision. With the primary scan



$$k(q) = b + Dq^2, \qquad k = 1/\tau_s.$$

Regression	D (10^{-3} m^2/s)	b (ps^{-1})	χ^2_ν (ν)
Finite-q TG	3.71 ± 0.70	0.0612 ± 0.0095	0.815 ($\nu = 2$)
Including PP	1.2 ± 1.6	0.099 ± 0.019	177 ($\nu = 3$)
TG retaining scan 4	3.78 ± 0.56	0.0594 ± 0.0076	0.553 ($\nu = 2$)

Parameter errors are regression standard errors; plotted rate bars are 95% repeat-scan bootstrap intervals. χ^2_ν uses bootstrap rate standard deviations, with unestimated diagnostic uncertainties (Appendix D.4).

Errors are conditional one-standard-error estimates; model uncertainty is excluded.

Figure 11: Slow-component rates against q^2: horizontal coordinates are calculated from the filename periods and the email's factor-of-four demagnification, while vertical coordinates are inverse slow lifetimes fitted to the recorded traces. Supplied delays and grating spacings are treated as exact. The finite-q line gives the TG-only diffusion estimate; including the $q = 0$ PP rate gives a poor linear fit. Bars show conditional 95% repeat-scan intervals for the rates; the dotted line shows the borderline-scan sensitivity.

masks held fixed, changing the fit start to 0.2, 1.0, or 2.0 ps gives (3.85 ± 1.02), (4.36 ± 1.25), or $(3.08 \pm 0.42) \times 10^{-3}$ m^2/s, respectively; the common factor 10^{-3} applies to all three values. The uncertainties in these sensitivity regressions are one-standard-error slope estimates, conditional on each selected window and scan set. This window dependence is distinct from the conditional bootstrap uncertainty.

4.4.1 Comparison of population and grating decay models

Section IV of Chen et al. [2] derives separate signals for grating imaging, total heterodyne detection and direct diffraction. Their weak-excitation total heterodyne expression, Eq. (4), has the structure

$$H(t) \propto F(t)\left[1 + \frac{1}{2}e^{-D_a q^2 t}\right], \qquad F(t) = A_r e^{-t/\tau_r} + A_c e^{-t/\tau_c}. \tag{89}$$

The non-diffusing term arises from the population response present in the total detected signal. The paper uses a GaAs/AlAs superlattice and a specific transmission geometry; neither its numerical lifetimes nor its fixed mixing coefficient can be transferred directly to our reflection experiment.

Under the tutorial's interpretation of the supplied TG traces as phase-extracted grating amplitudes, the appropriate quantity is the phase-dependent coefficient, rather than the whole signal in Eq. (89). This distinction follows explicitly by writing

$$S(t,\psi) - B = P(t) + G(t)\cos\psi, \tag{90}$$

$$\frac{S(t,\psi) + S(t,\psi+\pi)}{2} - B = P(t), \tag{91}$$

$$\frac{S(t,\psi) - S(t,\psi+\pi)}{2\cos\psi} = G(t) \quad (\cos\psi \neq 0). \tag{92}$$

The last equation isolates the grating response; it removes the additive population term. The raw phase scans and extraction metadata are unavailable, so this channel interpretation remains an experimental assumption. Adding the paper's non-diffusing term to already extracted TG amplitudes without checking their definition would mix distinct observables.

For a common diffusivity acting on the two components, the isolated response would be

$$G(t) = F(t)e^{-D_a q^2 t} \tag{93}$$

$$= A_r e^{-(1/\tau_r + D_a q^2)t} + A_c e^{-(1/\tau_c + D_a q^2)t}, \tag{94}$$

$$k_r(q) = 1/\tau_r + D_a q^2, \qquad k_c(q) = 1/\tau_c + D_a q^2. \tag{95}$$

Both rate branches must therefore have the same slope against q^2. Applying the existing analysis to both fitted branches gives

$$D_{\text{fast}}^{\text{eff}} = (2.05 \pm 0.63) \times 10^{-2} \text{ m}^2/\text{s}, \tag{96}$$

$$D_{\text{slow}}^{\text{eff}} = (3.71 \pm 0.70) \times 10^{-3} \text{ m}^2/\text{s}, \tag{97}$$

where the uncertainties are regression standard errors. These are diagnostic slopes, not two independently established material diffusivities. A paired 300-replicate bootstrap resamples whole retained scans and refits both components together, preserving their correlation. Its conditional 95% interval for $D_{\text{fast}}^{\text{eff}} - D_{\text{slow}}^{\text{eff}}$ is $[0.72, 2.72] \times 10^{-2}$ m^2/s. Thus the simple common-diffusivity assignment is inconsistent with the independently fitted components within the retained-scan variability. Model, fit-window and channel-definition uncertainties are excluded; the supplied delays and grating spacings are treated as exact. Figure 14 shows the two rate branches. The slow-branch result remains an exploratory effective estimate rather than a validated common transport coefficient.

The population response must also be described adequately before it can be compared with grating transport. Since the PP residuals show a long-lived contribution, we repeat the fit with three exponentials and a constant,

$$S_{\mathrm{PP}}^{(3)}(t) = B + \sum_{j=1}^{3} A_j e^{-t/\tau_j}, \qquad 0 < \tau_1 < \tau_2 < \tau_3, \tag{98}$$

using the same interval and scans. This gives (0.674 ± 0.011) ps, (5.22 ± 0.17) ps, and (619 ± 60) ps, with conditional covariance standard errors. The normalized residual RMS falls from 1.26% to 0.303%; uncertainties on these diagnostics have not been estimated. As shown in Fig. 14, the extra term accounts for much of the long-time trend, although structure remains. A slow instrumental background or a non-exponential response could also produce this improvement, so the longest fitted time cannot be assigned directly to recombination. The change in the fast time also affects its interpretation as cooling. We use the two-exponential fits for the main comparison and the three-exponential fit to show the dependence on the chosen model. This calculation is implemented in `analysis/paper_model_comparison.py`.

5 Discussion

The autocorrelation results show a shorter pulse after the compressor, as expected. Both fitted durations remain above the Gaussian transform limit obtained from the spectrum. In calculating these durations, the two overlap terms must be kept distinct and the changing fringe spacing must be included in the delay conversion. The remaining residual structure and dependence on the fitting interval show that the Gaussian fit does not fully describe the recorded signal. Measuring the mirror trajectory independently and using a phase-sensitive pulse measurement would help distinguish scan nonlinearity, chirp, and detector response.

For the carrier measurements, the main difficulty is assigning an optical decay time to a physical process. The fast PP component may describe cooling if the reflectivity change follows carrier temperature. This proportionality has not been measured, and the fitted time changes with the chosen interval and number of exponential terms. We can therefore compare its timescale with cooling, but cannot identify it uniquely with a temperature-relaxation time. Thermalization, individual phonon scattering events, cooling, and recombination describe different processes and need not have the same characteristic time.

The main transport result is the finite-q TG slope shown in Fig. 11. The fit yields

$$D_a = (3.71 \pm 0.70) \times 10^{-3}\,\mathrm{m^2\,s^{-1}} \quad = (37.1 \pm 7.0)\,\mathrm{cm^2\,s^{-1}}.$$

The repeat-scan bootstrap interval is $[2.31, 4.96] \times 10^{-3}\,\mathrm{m^2\,s^{-1}}$. A bulk-GaAs pump–probe study reports an ambipolar diffusivity near $20\,\mathrm{cm^2\,s^{-1}}$ at room temperature (approximate literature value; uncertainty not specified in the cited abstract), so the present estimate is of the same order [5]. This is not a controlled like-for-like comparison because our carrier density, fluence and measured sample temperature are unavailable; density-dependent effects can also change GaAs diffusivity [7].

The finite-q regression intercept corresponds to a nominal recombination time of (16.3 ± 2.5) ps (first-order propagation of the regression intercept standard error). The pump–probe slow lifetime is 7.878 ps with a conditional 95% repeat-scan interval $[7.753, 8.040]$ ps. Equivalently, its rate is $0.1269\ \mathrm{ps^{-1}}$ with a 95% interval $[0.1244, 0.1290]\ \mathrm{ps^{-1}}$, compared with the TG intercept $(0.0612 \pm 0.0095)\ \mathrm{ps^{-1}}$ (regression standard error); including the $q = 0$ point reduces the linear-fit R^2 to 0.161 (diagnostic uncertainty not estimated). This disagreement is the central limitation of the pump–probe comparison: the $q = 0$ reflectivity signal and finite-q diffracted signal may be sensitive to different mixtures of relaxation processes, while the simple rate law assumes one

common population and one recombination intercept. Thus the TG-only slope is an exploratory estimate, and the intercept should not be reported as a measured recombination lifetime.

Here, relaxation processes are physical mechanisms through which the excited sample returns toward equilibrium. Carrier cooling transfers excess carrier energy to the lattice; recombination reduces the number of excited electron–hole pairs; and diffusion redistributes carriers from densely excited regions into less excited regions, reducing the TG modulation without itself removing carriers. These processes can all affect the measured optical response, with different relative contributions to PP reflectivity and TG diffraction. Consequently, the fitted slow PP and TG components may approximate different mixtures of processes. Calling both components "slow" does not identify them with the same recombination mechanism. The two exponential terms are phenomenological descriptions of the traces; fitting them does not establish exactly two independent microscopic relaxation processes or uniquely assign a physical mechanism to either fitted lifetime.

Using $\mu_a = eD_a/(k_BT)$ at an assumed $T = 300\,\mathrm{K}$ gives a conditional mobility of $(1.44 \pm 0.27) \times 10^3\,\mathrm{cm^2\,V^{-1}\,s^{-1}}$. The mobility uncertainty propagates only the regression slope error at fixed assumed temperature; the corresponding 95% repeat-scan interval is $[0.89, 1.92] \times 10^3$ $\mathrm{cm^2/(V\,s)}$. The nominal condition is room temperature, but the actual sample temperature and its uncertainty were not logged, and the Einstein conversion assumes non-degenerate carriers in thermal equilibrium at a common temperature. Therefore this mobility is conditional on assumptions rather than a direct measurement. For the carrier analysis, the main unresolved issues are the fit interval and biexponential model, limited sampling of grating periods, scan-selection sensitivity, and missing acquisition metadata. Delay and grating-spacing calibration are treated as exact inputs rather than possible explanations of the PP/TG mismatch.

6 Conclusion

In this experiment, we characterized the laser pulses and studied the decay of photoexcited carriers in GaAs. The fitted pulse duration decreased from (112.89 ± 0.10) fs before compression to (93.902 ± 0.054) fs afterwards. Both values are above the spectral Gaussian transform limit of (83.87 ± 0.24) fs. The quoted errors are formal standard errors, while changes with the fitting interval show a larger uncertainty associated with the pulse model.

The PP and TG signals were described by fast and slow exponential decays. The fast PP time of 0.717 ps, with a 95% repeat-scan interval of $[0.698, 0.743]$ ps, gives an effective cooling-time estimate if the optical response follows carrier temperature. Without that calibration and the carrier heat capacity, an electron–phonon coupling coefficient cannot be determined.

The slow TG rates increase with q^2, giving an effective diffusivity of $(3.71 \pm 0.70) \times 10^{-3}$ $\mathrm{m^2/s}$. At an assumed 300 K, this corresponds to a mobility of $(1.44 \pm 0.27) \times 10^3$ $\mathrm{cm^2/(V\,s)}$. These errors follow from the regression at fixed supplied delays and grating spacings; model uncertainty and the unavailable temperature uncertainty are excluded. The PP rate does not agree with the TG intercept, and the independently fitted fast and slow TG components give different slopes. Thus the measurements support a dependence on grating spacing, but do not establish one common diffusion coefficient and recombination lifetime for all fitted components.

A Autocorrelation calibration and residuals

Figure 12 documents the residual structure of the selected raw-signal fits. The pulse-width analysis in Part I uses these residuals to assess the calibration and fit quality.

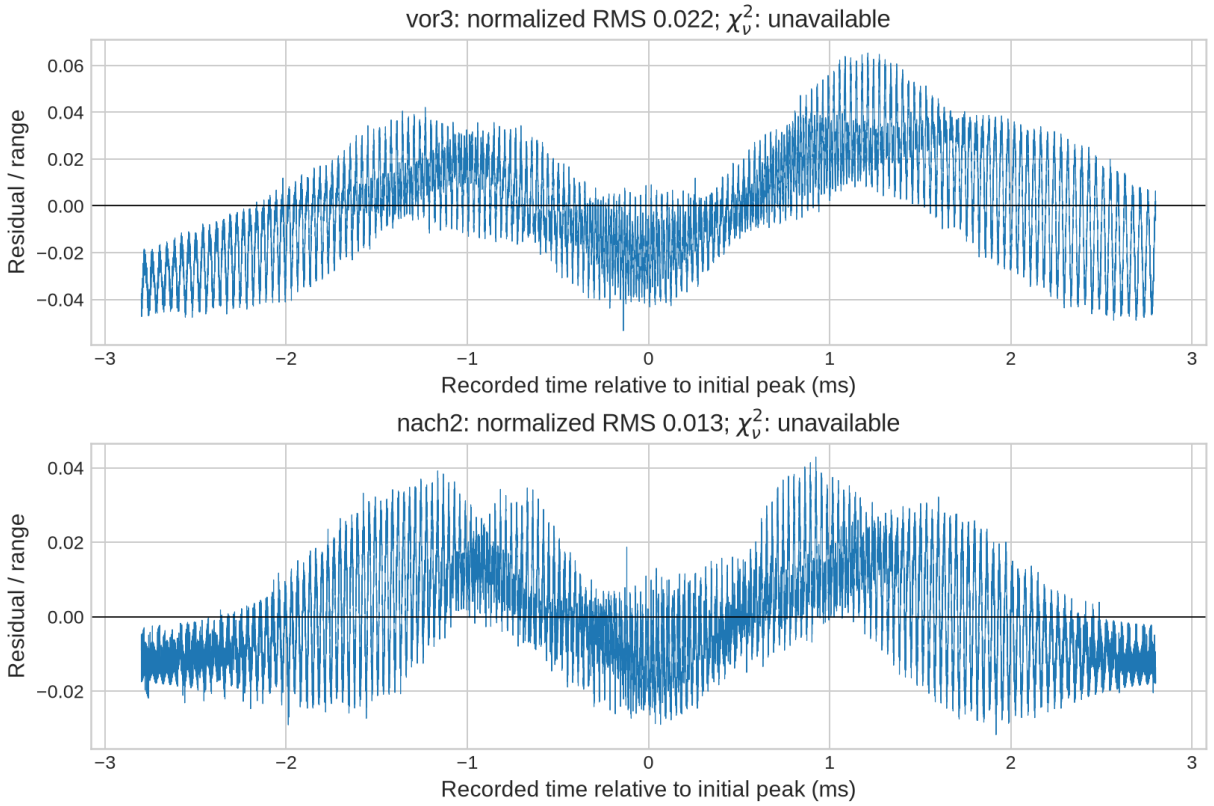
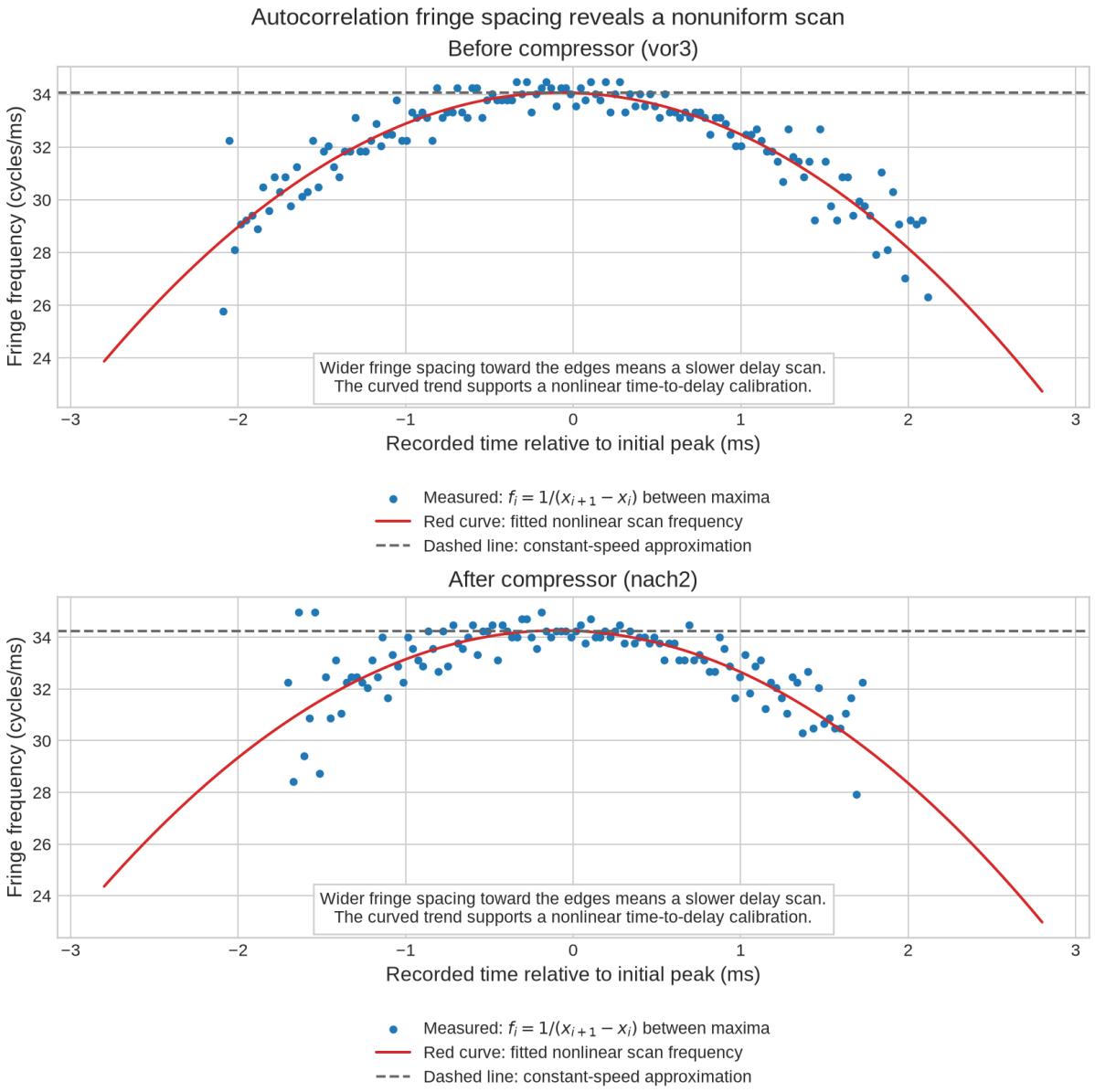


Figure 12: Selected autocorrelation fit residuals, normalized by the measured range in the fitted interval. Structured residuals remain after nonlinear fringe calibration and indicate that covariance errors alone understate uncertainty. RMS diagnostic uncertainties are unestimated; χ^2_ν is unavailable without pointwise measurement errors (Appendix D.4).



$$f_{\mathrm{fit}}(x) = du/dx = f + 2b_2(x - x_0) + 3b_3(x - x_0)^2.$$

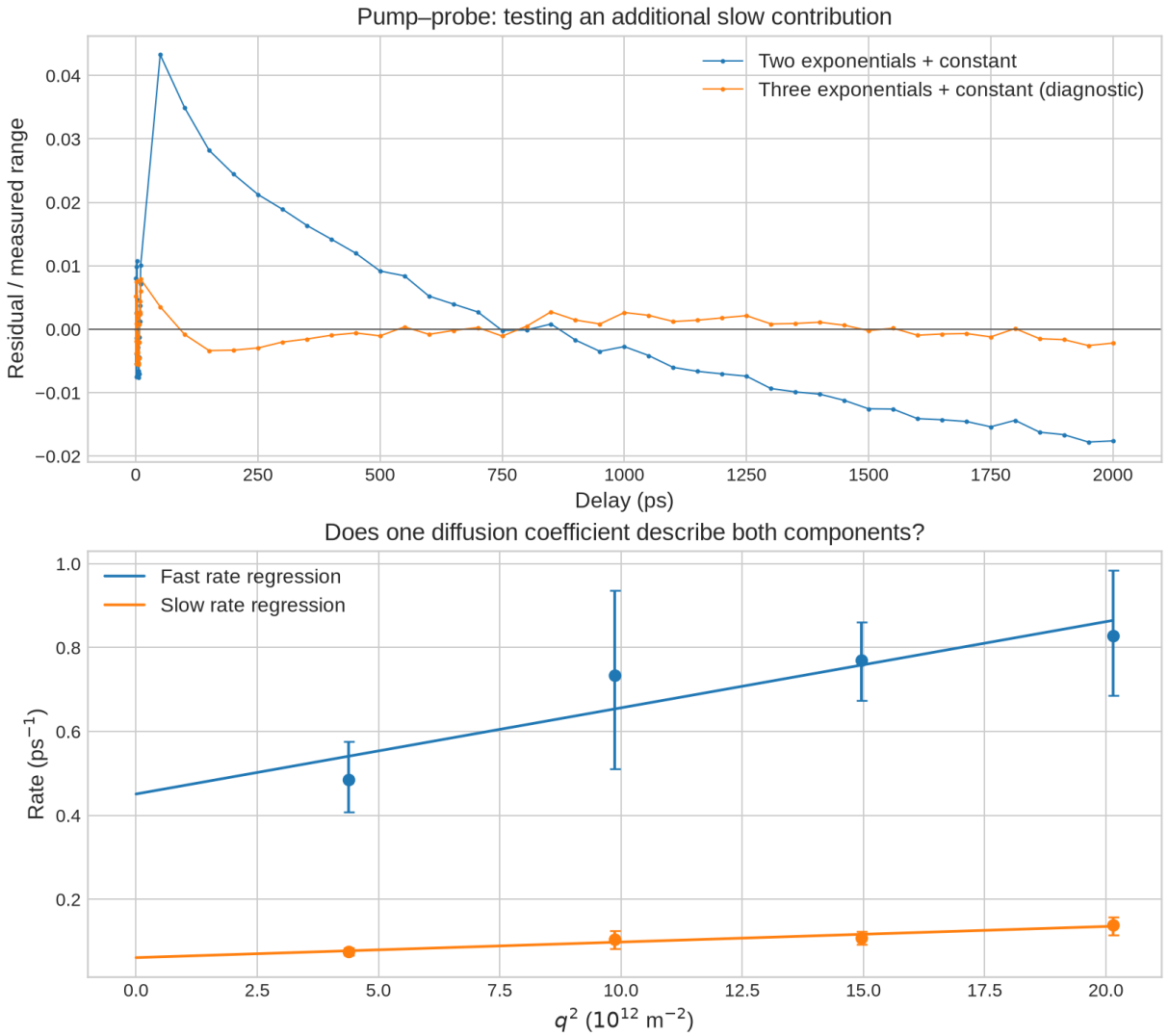
Parameter	vor3	nach2
x_0 (ms)	0.00869 ± 0.00072	0.00247 ± 0.00038
f (cycles/ms)	34.05357 ± 0.00031	34.25480 ± 0.00023
b_2 (cycles/ms^2)	-0.1140 ± 0.0010	-0.12715 ± 0.00055
b_3 (cycles/ms^3)	-0.45726 ± 0.00016	-0.45029 ± 0.00019

The red curves are derivatives of the raw-signal fits, rather than independent fits to the blue points, so no separate χ^2_ν is assigned. The dashed lines are f.

Errors are conditional one-standard-error estimates; model uncertainty is excluded.

Figure 13: Recorded fringe rate: blue points are inverse separations between successive measured maxima, placed at interval midpoints; red curves give the fitted delay-map derivative and dashed lines the central rate. Wider fringe spacing toward the edges supports a nonuniform delay scan. This is a recorded fringe rate, distinct from the optical carrier frequency. Pointwise uncertainties are unquantified; peak selection is explained in Sec. 2.4.

B Population and transport model diagnostics



$$S_m(t) = C + \sum_{j=1}^{m} A_j e^{-t/\tau_j}, \quad m = 2, 3; \qquad k_{f,s}(q) = b_{f,s} + D_{f,s} q^2.$$

Parameter	PP two-exp. / fast rate	PP three-exp. / slow rate
τ_1 (ps)	0.717 ± 0.042	0.674 ± 0.011
τ_2 (ps)	7.88 ± 0.79	5.22 ± 0.17
τ_3 (ps)	–	619 ± 60
A_1	182.9 ± 6.5	181.5 ± 1.8
A_2	42.3 ± 3.2	42.6 ± 1.1
A_3	–	8.47 ± 0.28
C	35.22 ± 0.28	32.82 ± 0.23
χ^2_ν (PP)	120 ($\nu = 59$)	1.98 ($\nu = 57$)
D_f, D_s (10^{-3} m^2/s)	20.5 ± 6.3	3.71 ± 0.70
b_f, b_s (ps^{-1})	0.451 ± 0.086	0.0612 ± 0.0095
χ^2_ν (rate lines)	1.07 ($\nu = 2$)	0.815 ($\nu = 2$)

A_j, C: 10^{-6} recorded units. PP errors: covariance SE; rate-line errors: regression SE. χ^2_ν values are diagonal diagnostics with unestimated uncertainties (Appendix D.4). The last three rows describe the fast/slow rate lines.

Errors are conditional one-standard-error estimates; model uncertainty is excluded.

Figure 14: Checks motivated by Chen et al. Top: PP residuals for two and three exponentials plus a constant. Bottom: independent fast- and slow-rate regressions; bars show conditional 95% whole-scan bootstrap intervals. Equal slopes would follow from a shared diffusivity. Supplied delays and grating spacings are treated as exact.

C Answers to the tutorial questions

The following answers address the preparation questions, calculations, and bonus question in the tutorial [3], in their original order. They complement the theory and analysis in the main text. Constants obtained from the ideal models are given as exact expressions or rounded values and do not carry measurement uncertainties.

C.1 Preparation: optical components and laser operation

What do the optical components do, and how does a solid-state laser work? A prism refracts different wavelengths through different angles and can therefore separate spectral components or provide adjustable dispersion in a compressor. A waveplate introduces a phase difference between two orthogonal polarization components: a half-wave plate rotates linear polarization, while a quarter-wave plate can convert appropriately oriented linear polarization into circular polarization. A polarizer selects one polarization component. The BBO crystal converts part of the fundamental light into its second harmonic.

LASER stands for *Light Amplification by Stimulated Emission of Radiation*. A solid-state laser contains a pumped gain medium, a resonator with mirrors, and a partially transmitting output coupler. Pump absorption establishes a population inversion on the laser transition. A resonant photon can stimulate an excited ion to emit into the same optical mode; repeated passes amplify the light when the round-trip gain exceeds the losses. Gain saturation sets the steady output. The broad gain bandwidth of Ti:sapphire supports many longitudinal modes. Their frequencies satisfy the cavity resonance condition; for an ideal empty linear cavity of length L, their separation is $c/(2L)$.

For equally spaced modes with locked phases, the interference is explicitly

$$E(t) = E_0 e^{i\omega_0 t} \sum_{m=0}^{M-1} e^{im\Delta\omega t} = E_0 e^{i\omega_0 t} \frac{1 - e^{iM\Delta\omega t}}{1 - e^{i\Delta\omega t}}, \tag{99}$$

$$|E(t)|^2 = |E_0|^2 \frac{\sin^2(M\Delta\omega t/2)}{\sin^2(\Delta\omega t/2)}. \tag{100}$$

The limiting peak intensity is $M^2|E_0|^2$ at multiples of $2\pi/\Delta\omega$; adding locked modes narrows each peak. In Kerr-lens mode locking, $n = n_0 + n_2 I$ makes the intense beam center focus more strongly. An aperture or the spatial overlap with the pumped gain region then favors intense pulses through lower effective loss. Dispersion compensation helps maintain their short duration.

C.2 Nonlinear optics and the role of BBO

Why is BBO used, and what does a nonlinear optical response mean? At sufficiently strong fields, the induced polarization is not simply proportional to the applied field. Suppressing tensor indices,

$$P = \epsilon_0\big(\chi^{(1)}E + \chi^{(2)}E^2 + \cdots\big), \tag{101}$$

$$E(t) = E_0 \cos\omega t, \tag{102}$$

$$P^{(2)}(t) = \frac{\epsilon_0 \chi^{(2)} E_0^2}{2}\big[1 + \cos(2\omega t)\big]. \tag{103}$$

The quadratic term oscillates at twice the incident frequency and produces second-harmonic light. BBO is suitable because its non-centrosymmetric structure permits a second-order response and its orientation can be adjusted for phase matching. For weak conversion and sufficient bandwidth, the field satisfies $E_{2\omega} \propto E_\omega^2$, giving $I_{2\omega} \propto I_\omega^2$ for the intensity. The BBO crystal and filter in Fig. 3 allow us to detect this nonlinear overlap signal.

C.3 Gaussian and Lorentzian time–bandwidth products

What are the transform-limited constants for the two pulse shapes? Both widths below are intensity FWHM values, and frequency is measured in cycles per second. For the Gaussian field $a(t) = e^{-t^2/\Gamma^2}$, the Gaussian integration derived in the theoretical background gives

$$|a(t)|^2 = e^{-2t^2/\Gamma^2}, \qquad |\widetilde{a}(\nu)|^2 = \pi\Gamma^2 e^{-2\pi^2\Gamma^2\nu^2}, \tag{104}$$

$$\Delta t = \Gamma\sqrt{2\ln 2}, \qquad \Delta\nu = \frac{\sqrt{2\ln 2}}{\pi\Gamma}, \tag{105}$$

$$K_{\mathrm{G}} = \Delta t\,\Delta\nu = \frac{2\ln 2}{\pi} \simeq 0.441. \tag{106}$$

For the conventional *Lorentzian field envelope*, $a(t) = 1/[1 + (t/a_0)^2]$, with $a_0 > 0$, the intensity is a squared Lorentzian. To evaluate its Fourier integral, set $\Omega = 2\pi\nu$:

$$F(\Omega) = \int_{-\infty}^{\infty} \frac{a_0^2 e^{-i\Omega t}}{(t - ia_0)(t + ia_0)}\, dt. \tag{107}$$

For $\Omega > 0$, close the contour in the lower half-plane, where the exponential decays. The contour is clockwise and encloses the pole at $-ia_0$; its semicircle contribution vanishes. Hence

$$\mathrm{Res}_{z=-ia_0} \frac{a_0^2 e^{-i\Omega z}}{(z - ia_0)(z + ia_0)} = \frac{a_0^2 e^{-a_0\Omega}}{-2ia_0}, \tag{108}$$

$$F(\Omega) = -2\pi i \frac{a_0^2 e^{-a_0\Omega}}{-2ia_0} = \pi a_0 e^{-a_0\Omega}. \tag{109}$$

For $\Omega < 0$, closing in the upper half-plane gives $\pi a_0 e^{a_0\Omega}$; at zero, direct integration gives $a_0[\arctan(t/a_0)]_{-\infty}^{\infty} = \pi a_0$. Thus $F(\Omega) = \pi a_0 e^{-a_0|\Omega|}$. The half-maximum conditions then give

$$\left[1 + (t/a_0)^2\right]^{-2} = \tfrac{1}{2} \quad \Rightarrow \quad \Delta t = 2a_0\sqrt{\sqrt{2} - 1}, \tag{110}$$

$$e^{-4\pi a_0|\nu|} = \tfrac{1}{2} \quad \Rightarrow \quad \Delta\nu = \frac{\ln 2}{2\pi a_0}, \tag{111}$$

$$K_{\mathrm{L}} = \frac{\ln 2}{\pi}\sqrt{\sqrt{2} - 1} \simeq 0.142. \tag{112}$$

These conventions agree with the pulse-shape table in Kärtner's lecture notes [1]. Calling the *intensity* Lorentzian would specify a different pulse and would not justify using this constant. The tutorial does not specify this distinction.

C.4 Benefits, limitations, and stretching of ultrashort pulses

What does the time–bandwidth product imply for an experiment? Short pulses resolve fast changes and provide high peak intensity at a given pulse energy. Their necessarily broad spectrum limits spectral selectivity and may excite several transitions. High peak intensity can also cause unwanted nonlinear responses or damage, while finite pump and probe durations broaden the measured temporal response. The relevant resolution is their cross-correlation at the sample.

A transform-limited pulse has no spectral phase curvature. Writing the spectral phase near ω_0 as

$$\phi(\omega) = \phi_0 + \phi_1(\omega - \omega_0) + \tfrac{1}{2}\phi_2(\omega - \omega_0)^2 + \cdots, \tag{113}$$

$$t_g(\omega) = \frac{d\phi}{d\omega} = \phi_1 + \phi_2(\omega - \omega_0) + \cdots \tag{114}$$

shows that nonzero ϕ_2 gives different frequency components different arrival times. Glass, the gain crystal, and other optics can introduce this dispersion. Nonlinear phase modulation and imperfect compression can also alter the pulse. An enlarged measured time–bandwidth product can additionally reflect an incorrect shape assumption or calibration. Figure 6 supplies the bandwidth and Fig. 7 the pulse-duration comparison; neither alone reconstructs spectral phase.

C.5 Autocorrelator alignment and second-harmonic detection

Why must the replicas overlap in space and time, and why detect the second harmonic? The product $E(t)E(t-\tau)$ is appreciable only where both fields are present. Their focused spots must coincide inside BBO, their polarizations must suit the nonlinear interaction, and their path lengths must bring them within a pulse duration of each other. Steering mirrors set spatial overlap; the translation stage sets the mean delay and the vibrating mirror scans it. A mirror displacement Δx changes the Michelson round-trip delay by $2\Delta x/c$.

The time-integrated fundamental signal is

$$S_\omega(\tau) \propto 2\int |E(t)|^2 dt + 2\,\mathrm{Re}\int E(t)E^*(t-\tau)\,dt. \tag{115}$$

Its delay-dependent part is a field autocorrelation. Its Fourier transform gives the spectral intensity, so it cannot by itself determine pulse stretching caused by spectral phase. Second-harmonic detection instead measures

$$S_{2\omega}(\tau) \propto \int |E(t)+E(t-\tau)|^4 dt, \tag{116}$$

which contains the intensity-overlap integral. It converts femtosecond overlap into a slowly recorded signal as the delay is scanned. For identical replicas with ideal coherent overlap, zero delay gives $16\int|E|^4dt$ and separated pulses give $2\int|E|^4dt$, hence the ideal peak-to-background ratio is $8:1$. These are theoretical ratios, not measured contrasts.

C.6 The autocorrelation envelope and the Gaussian width parameter

Why can the envelope determine pulse duration, and what does the fitted Gaussian width mean? For a specified pulse shape and phase model, the overlap factors vary on the pulse-duration scale while the fringes vary on the optical-period scale. The envelope therefore constrains duration without requiring the absolute optical phase. This is a model-dependent inference: an autocorrelation does not uniquely reconstruct an arbitrary pulse. In this experiment the fringes also calibrate the nonlinear delay scan (Fig. 13), and the actual fit uses the complete trace in Eq. (77).

In the tutorial's expression $E(t) = e^{-t^2/\Gamma^2}$, Γ is a Gaussian scale parameter, *not* an FWHM. The two relevant half-maximum equations are

$$|E(t)| = \tfrac{1}{2} \Rightarrow |t| = \Gamma\sqrt{\ln 2} \qquad \Rightarrow \Delta t_{\text{field}} = 2\Gamma\sqrt{\ln 2}, \tag{117}$$

$$I(t) = e^{-2t^2/\Gamma^2} = \tfrac{1}{2} \Rightarrow |t| = \Gamma\sqrt{\ln 2/2} \qquad \Rightarrow \Delta t_{\text{pulse}} = \Gamma\sqrt{2\ln 2}. \tag{118}$$

For the isolated intensity autocorrelation, completing the square gives

$$G(\tau) = \int_{-\infty}^{\infty} e^{-2[t^2+(t-\tau)^2]/\Gamma^2}\,dt \tag{119}$$

$$= e^{-\tau^2/\Gamma^2}\int_{-\infty}^{\infty} e^{-4(t-\tau/2)^2/\Gamma^2}\,dt \tag{120}$$

$$= \frac{\Gamma}{2}e^{-\tau^2/\Gamma^2}\int_{-\infty}^{\infty} e^{-u^2}\,du = \frac{\sqrt{\pi}\Gamma}{2}e^{-\tau^2/\Gamma^2}, \tag{121}$$

where $u = 2(t - \tau/2)/\Gamma$. Consequently,

$$\Delta t_{\mathrm{AC}} = 2\Gamma\sqrt{\ln 2} = \sqrt{2}\,\Delta t_{\mathrm{pulse}}. \tag{122}$$

The photodiode actually records integrated second-harmonic energy versus delay. The full interferometric envelope contains several overlap terms with different widths, as Eq. (65) shows. Dividing its entire envelope FWHM by $\sqrt{2}$ is therefore unjustified; that factor applies to $G(\tau)$. The report quotes the underlying fitted *intensity* pulse FWHM.

C.7 Calculation tasks: diffusion solution and Fourier transformation

Does the proposed density solve the diffusion equation? Define $k_0 = 1/\tau_{\mathrm{rec}}$ and $k_q = k_0 + D_a q^2$. For the proposed density,

$$n(x,t) = n_0[e^{-k_0 t} + \cos(qx)e^{-k_q t}], \tag{123}$$

$$\partial_t n = -n_0[k_0 e^{-k_0 t} + k_q \cos(qx)e^{-k_q t}], \tag{124}$$

$$\partial_x^2 n = -n_0 q^2 \cos(qx)e^{-k_q t}, \tag{125}$$

$$D_a\partial_x^2 n - k_0 n = -n_0[k_0 e^{-k_0 t} + (D_a q^2 + k_0)\cos(qx)e^{-k_q t}] = \partial_t n. \tag{126}$$

It also has the required initial condition $n(x,0) = n_0[1 + \cos(qx)]$.

How does the Fourier transformation produce tutorial Eq. (4.4)? To handle the ideal periodic grating without integrating a non-decaying density over the entire real line, choose a length L containing an integer number of periods. Define

$$\widetilde{n}_k(t) = \frac{1}{L}\int_0^L n(x,t)e^{-ikx}\,dx, \qquad k = \frac{2\pi m}{L}, \quad m \in \mathbb{Z}. \tag{127}$$

Two integrations by parts yield

$$\frac{1}{L}\int_0^L \partial_x^2 n\, e^{-ikx}\,dx = \frac{1}{L}[\partial_x n\, e^{-ikx}]_0^L + \frac{ik}{L}\int_0^L \partial_x n\, e^{-ikx}\,dx \tag{128}$$

$$= \frac{1}{L}[\partial_x n\, e^{-ikx}]_0^L + \frac{ik}{L}[n\, e^{-ikx}]_0^L - \frac{k^2}{L}\int_0^L n\, e^{-ikx}\,dx \tag{129}$$

$$= -k^2\widetilde{n}_k. \tag{130}$$

The boundary terms vanish by periodicity. Transforming both sides of the diffusion equation therefore gives

$$\frac{d\widetilde{n}_k}{dt} = -(D_a k^2 + k_0)\widetilde{n}_k, \tag{131}$$

$$\int_{\widetilde{n}_k(0)}^{\widetilde{n}_k(t)} \frac{d\widetilde{n}_k}{\widetilde{n}_k} = -(D_a k^2 + k_0)\int_0^t dt', \tag{132}$$

$$\widetilde{n}_k(t) = \widetilde{n}_k(0)e^{-(D_a k^2 + k_0)t}. \tag{133}$$

Initially only the coefficients at $k = 0, \pm q$ are nonzero: they are $n_0, n_0/2, n_0/2$. Substitution into the inverse Fourier series recovers the proposed solution above. The same $-k^2$ rule holds for the Fourier integral of a localized density when the boundary terms vanish at infinity.

C.8 Finite wave number, uniform excitation, and diffusion measurements

What are the lifetimes for finite q and for $q = 0$, and how is D_a obtained? The grating-amplitude lifetime is

$$\tau_q = \frac{1}{1/\tau_{\mathrm{rec}} + D_a q^2}, \qquad \tau_{q=0} = \tau_{\mathrm{rec}}. \tag{134}$$

The $q = 0$ component is spatially uniform and has no lateral density gradient in this ideal model. A pump–probe population measurement approximates this case when lateral diffusion out of the probed region is negligible. A real finite spot or a depth-dependent population can introduce additional transport.

Measure TG amplitude decays for several calibrated periods, calculate $q = 2\pi/\Lambda_{\mathrm{sample}}$, and regress the decay rate against q^2. The slope gives D_a and the intercept gives $1/\tau_{\mathrm{rec}}$ if all traces probe the same population. Alternatively, a matched population trace and one TG trace give

$$D_a = \frac{k_{\mathrm{TG}}(q) - k_{\mathrm{pop}}}{q^2}. \tag{135}$$

Two finite wave numbers also suffice algebraically:

$$D_a = \frac{k(q_1) - k(q_2)}{q_1^2 - q_2^2}, \qquad k_0 = k(q_1) - D_a q_1^2, \tag{136}$$

but additional periods allow the assumed linear dependence to be checked. For the supplied data the sample period is one quarter of the mask period. Figure 11 shows why the measured PP rate cannot simply be used as the common intercept.

C.9 Bonus: other optically generated gratings

Which gratings can be created besides a carrier-density grating? An optical grating can also be formed by heating the sample periodically, which changes its temperature and refractive index. Periodic stress can launch acoustic waves and create a strain grating. With suitable optical selection rules, a spatially varying pump polarization can produce a spin or polarization grating even at constant total intensity. Such measurements can be used to study heat transport, acoustic propagation, and spin dynamics. These are possible extensions of the technique; the present data do not establish these additional signals.

C.10 Heterodyne sensitivity and separation of population and grating signals

Why use heterodyne detection, and how are the two contributions separated? The detector measures

$$|E_{\mathrm{LO}} + E_{\mathrm{dif}}|^2 = |E_{\mathrm{LO}}|^2 + |E_{\mathrm{dif}}|^2 + 2\,\mathrm{Re}(E_{\mathrm{LO}}^* E_{\mathrm{dif}}). \tag{137}$$

For a weak diffracted field, mixing with a stronger local oscillator produces a term linear in E_{dif} that can exceed the directly detected $|E_{\mathrm{dif}}|^2$ term. It also retains phase information. Increasing the local oscillator is useful within detector and noise limits; it does not imply unlimited signal-to-noise improvement.

At each pump–probe delay, scan the coverslip angle and fit the phase dependence shown in Fig. 4. In the tutorial's model,

$$S_{\mathrm{Het}}(t, \psi) = B + S_{\mathrm{Pop}}(t) + S_{\mathrm{TG}}(t) \cos\psi, \tag{138}$$

$$S_{\mathrm{Pop}}(t) = \frac{S_{\mathrm{Het}}(t, 0) + S_{\mathrm{Het}}(t, \pi)}{2} - B, \tag{139}$$

$$S_{\mathrm{TG}}(t) = \frac{S_{\mathrm{Het}}(t, 0) - S_{\mathrm{Het}}(t, \pi)}{2}. \tag{140}$$

Here B is a separately established instrumental offset. A full phase scan can fit an unknown phase origin as well. Opposite-phase subtraction isolates the grating term; averaging isolates the phase-independent population contribution. Pump chopping alone removes the unmodulated background but does not generally remove the pump-induced population term.

The tutorial calls single-probe transient reflectivity homodyne and the added-reference TG measurement heterodyne. These names describe its detection arrangement; using a common source for pump and probe does not decide the classification. Indeed,

$$\Delta I_{\mathrm{PP}} = |E_{\mathrm{probe}}|^2\big(|r_0+\delta r|^2 - |r_0|^2\big) \simeq 2|E_{\mathrm{probe}}|^2\,\mathrm{Re}(r_0^*\delta r), \tag{141}$$

so the reflected probe supplies an internal reference for the induced field change. In terminology that reserves heterodyne for a frequency-shifted reference, same-frequency mixing is called homodyne; the tutorial uses the broader spectroscopy convention.

How are recombination and diffusion rates separated experimentally? After extracting the two phase components, fit their time dependence and use $k_{\mathrm{TG}} - k_{\mathrm{Pop}} = D_a q^2$, or vary the grating period and determine slope and intercept as above. If $E_{\mathrm{dif}} \propto n_q$, direct diffracted intensity instead decays as $|n_q|^2 \propto e^{-2k_q t}$; its measured rate must be divided by two before applying the amplitude-rate law. These procedures require matched physical components. The two empirical biexponential times in Table 2 do not by themselves establish two microscopic populations or a unique recombination time.

C.11 Chopper, lock-in amplifier, and prism compressor

How do modulation and compression help the measurement? A rotating slotted chopper periodically blocks the pump. The pump-induced probe response is then modulated at a known reference frequency. A lock-in multiplies the detector signal by synchronized sine and cosine references and averages over many cycles. For $s(t) = A\cos(\Omega t+\varphi)+B$, an averaging interval containing complete cycles gives

$$X = 2\langle s(t)\cos\Omega t\rangle = A\cos\varphi, \tag{142}$$

$$Y = -2\langle s(t)\sin\Omega t\rangle = A\sin\varphi, \tag{143}$$

because $\langle\cos^2\Omega t\rangle = \langle\sin^2\Omega t\rangle = 1/2$ and the mixed and DC-reference averages vanish. Thus $\sqrt{X^2+Y^2} = |A|$. A real chopped signal contains harmonics; the lock-in selects the component at its chosen reference frequency. This suppresses DC backgrounds and noise outside the detection bandwidth, with a tradeoff between averaging time and acquisition speed.

A prism compressor gives different wavelengths adjustable optical path lengths. Properly chosen prism spacing and insertion compensate accumulated dispersion so that spectral components overlap more closely in time. It improves time resolution and peak intensity at fixed pulse energy; it does not require a narrower spectrum. Residual dispersion, losses, and alignment sensitivity limit the achieved compression. The compressor location and modulation arrangement are shown in Fig. 2.

C.12 Ambipolar mobility limits and the band structure

What do the equal- and unequal-mobility limits mean? Starting from the tutorial's expression,

$$\mu_a = \frac{2\mu_e\mu_h}{\mu_e+\mu_h}, \tag{144}$$

$$\mu_e = \mu_h = \mu \Rightarrow \mu_a = \mu, \tag{145}$$

$$\mu_j \gg \mu_i \Rightarrow \mu_a = \frac{2\mu_i}{1+\mu_i/\mu_j} \longrightarrow 2\mu_i. \tag{146}$$

Equal mobilities give the same ambipolar mobility. For strongly unequal mobilities, the internal electric field couples the two species and the slower species limits the joint transport. The printed second limit in tutorial Eq. (4.14) is a typographical error: it should state $\mu_a \simeq 2\mu_i$ for $\mu_j \gg \mu_i$.

Band curvature determines the effective-mass tensor:

$$(m^{*-1})_{\alpha\beta} = \frac{1}{\hbar^2}\frac{\partial^2 E}{\partial k_\alpha \partial k_\beta}. \tag{147}$$

For an isotropic parabolic band in the relaxation-time approximation, the steady drift follows from

$$0 = q_c\mathbf{E} - \frac{m^*\mathbf{v}_d}{\tau_m} \quad \Rightarrow \quad \mu = \frac{|\mathbf{v}_d|}{|\mathbf{E}|} = \frac{e\tau_m}{m^*}, \tag{148}$$

where $q_c = \pm e$ and τ_m is the momentum-relaxation time. Greater curvature means smaller effective mass and, at fixed τ_m, larger mobility. Scattering, band anisotropy, and occupation of different valleys also matter; band curvature alone does not determine the mobility.

C.13 Where the tutorial's analysis tasks are answered

Part I reports the fringe-based delay calibration, spectral bandwidth, pulse widths, and time–bandwidth comparison, with Figs. 6, 7, and 13. Part II gives the conditional cooling-time interpretation and derives the coupling relation; a volumetric electron–phonon coefficient remains undetermined because the necessary heat-capacity and excitation information is missing. Part III gives the transient-reflectivity and TG decay constants with confidence intervals in Table 2. Part IV and the discussion assess the effective diffusion coefficient, conditional mobility, literature comparison, and failure of a common recombination intercept. These answers retain the distinction between measured optical decay times and model-dependent material quantities.

D Statistical methods and interpretation

D.1 Bootstrapping

To estimate how repeat measurements affect a fitted result, we use bootstrapping. We repeatedly draw measurements from the available dataset *with replacement*. A measurement can therefore be selected more than once, while another may be left out. Each resample has the same size as the original dataset. Repeating the analysis for these resamples gives a distribution of fitted values, whose spread describes the repeat variability.

We resample complete retained scans. If a file contains n scans $S_1(t), \ldots, S_n(t)$ at the same delays, we draw n indices independently and uniformly from $\{1, \ldots, n\}$ for each resample r. Its mean is

$$J_1^{(r)}, \ldots, J_n^{(r)}, \qquad \overline{S}^{*(r)}(t) = \frac{1}{n} \sum_{j=1}^{n} S_{J_j^{(r)}}(t).$$

We fit this mean using the same function, interval, and procedure as for the original data. Repeating the calculation $R = 300$ times gives lifetime estimates $\tau^{*(1)}, \ldots, \tau^{*(R)}$. The value reported in the results is still the fit to the original mean; the resampled fits are used to estimate its uncertainty.

For quantities involving several files, scans are resampled separately within each file. Each replicate then repeats all relevant fits and derived calculations. In particular, slow lifetimes are converted to rates and regressed against the unchanged q^2 coordinates, treated as exact to obtain a bootstrap diffusion estimate. The paired fast/slow diagnostic uses the same resampled scans for both components, preserving their correlation.

Resampling whole scans preserves relationships between delay points within a scan. Treating all individual delay points as independent would discard that structure. The method nevertheless assumes that the retained scans adequately represent repeat measurements under comparable conditions; it cannot recover sources of variability absent from those scans.

D.2 Repeat-scan bootstrap intervals

A *repeat-scan bootstrap interval* summarizes the spread of the estimates obtained by the procedure above. The conditional 95% percentile interval for a parameter θ is

$$\mathrm{CI}_{95\%}(\theta) = \left[Q_{0.025}(\theta^*),\ Q_{0.975}(\theta^*) \right],$$

where Q_p denotes the p quantile of the bootstrap estimates. Its endpoints are the 2.5th and 97.5th percentiles, enclosing the middle 95% of those estimates. The interval need not be symmetric about the original estimate.

The quantiles are calculated by sorting the N bootstrap estimates as $x_0 \le x_1 \le \cdots \le x_{N-1}$ and applying NumPy's default linear interpolation. With indices starting at zero and $0 < p < 1$, define

$$h = (N-1)p, \qquad j = \lfloor h \rfloor, \qquad a = h - j.$$

The quantile is the interpolated value

$$Q_p = (1-a)x_j + a x_{j+1}.$$

An integer h gives $a = 0$ and selects a sorted estimate directly; otherwise the quantile lies between its two neighboring estimates. The implementation uses `np.percentile` with the requested percentages 2.5 and 97.5.

For the $N = 300$ replicates used here, the lower bound is

$$h = 299(0.025) = 7.475, \qquad Q_{0.025} = 0.525x_7 + 0.475x_8,$$

between the eighth and ninth smallest estimates. The upper bound is

$$h = 299(0.975) = 291.525, \qquad Q_{0.975} = 0.475x_{291} + 0.525x_{292},$$

between the 292nd and 293rd smallest estimates. These bounds are taken from the ordered bootstrap distribution, rather than calculated as the fitted value plus or minus a multiple of its standard deviation.

These intervals estimate uncertainty arising from variation among the retained repeated scans, conditional on the chosen model, fit window, and fixed scan-selection masks. The supplied PP/TG delays and grating spacings are assumed exact. The intervals do not include uncertainty in the excluded-scan selection, unrepresented acquisition drift, or an unsuitable physical model. A narrow interval can therefore coexist with systematic residuals. The interval is also different from a conditional covariance standard error written as $\pm u$: the two calculations describe different sources of statistical variability. A 95% confidence interval is not a statement that 95% of individual measurements lie inside its bounds, or a guaranteed probability statement about this particular realized interval.

Derived quantities are recalculated for each resample. For example, $k^{*(r)} = 1/\tau^{*(r)}$ supplies a bootstrap distribution of rates. Since the reciprocal is decreasing for positive lifetimes, lifetime bounds $[L, U]$ correspond to rate bounds $[1/U, 1/L]$. The plotted rate intervals are calculated directly from the resampled rates. Neither symmetric lifetime errors nor their reciprocals should be treated as automatically symmetric rate errors.

D.3 Coefficient of determination R^2

The coefficient of determination R^2 describes how much of the variation in the fitted decay rates is accounted for by a straight line. For rates $k_i = 1/\tau_{s,i}$, their mean $\overline{k}$, and the line predictions $\widehat{k}_i = b + D_a q_i^2$, it is defined as

$$R^2 = 1 - \frac{\sum_i (k_i - \widehat{k}_i)^2}{\sum_i (k_i - \overline{k})^2}.$$

The numerator measures the squared deviations from the fitted line; the denominator measures the squared deviations from the mean rate. For the unweighted least-squares regressions with a freely fitted intercept used here, $R^2 = 1$ means that all points lie on the line, whereas $R^2 = 0$ means that the line predicts no better than the constant mean. This interpretation requires nonzero variation in the rates.

The TG-only value $R^2 = 0.934$ corresponds to 93.4% of the rate variation accounted for by the line; including PP gives $R^2 = 0.161$, or 16.1%. These are descriptive statistics with unestimated uncertainties, calculated for four and five points, respectively. Their comparison is not a formal statistical test. A high R^2 alone does not validate the physical model, quantify parameter uncertainty, or account for model mismatch and systematic residuals. It also does not show that the fitted slow components represent the same microscopic process in every dataset.

D.4 Reduced chi-squared diagnostics

If a fitted curve $f(x_i; \widehat{\boldsymbol{\theta}})$ is compared with N observations y_i having standard errors s_i, a diagonal reduced chi-squared statistic is

$$\chi^2_\nu = \frac{1}{\nu} \sum_{i=1}^{N} \frac{[y_i - f(x_i; \widehat{\boldsymbol{\theta}})]^2}{s_i^2}, \qquad \nu = N - p,$$

where p is the number of independently fitted parameters. The nominal denominators use $p = 5$ for a biexponential plus constant, $p = 7$ for three exponentials plus constant, and $p = 2$ for a straight line. The numerical values are descriptive diagnostics; uncertainties on these statistics have not been estimated.

For PP and TG mean traces, s_i is the sample standard deviation across retained scans at delay t_i, divided by the square root of the number of retained scans. For rate regressions, s_i is the standard deviation of that rate's 300 whole-scan bootstrap estimates. It is not the half-width of its 95% percentile interval. The regression retaining borderline scan 4 uses a separate resampling calculation for that scan selection. The supplied delay and grating coordinates remain exact inputs.

The fits minimize unweighted squared residuals, while this statistic divides the residuals by estimated errors afterwards. In addition, delay points within a scan may be correlated, and the errors are estimated from a finite number of repeats. The diagonal sum neglects these correlations. We therefore use $N-p$ as a nominal number of degrees of freedom and do not assign a formal goodness-of-fit probability or rejection threshold. A large value indicates deviations that are large compared with the estimated repeat variability. It must be interpreted together with the residual plots; a value near unity alone does not validate the model.

The single spectrum and IAC records have no supplied pointwise measurement errors or independent noise model, so their reduced chi-squared is marked unavailable. This also applies to their fit-window variants. Setting a common noise variance equal to $\widehat{s}^2 = \sum_i [y_i - f(x_i)]^2/(N-p)$ would force $\chi^2_\nu = 1$ by construction and provide no independent check. The fringe-frequency curves are derivatives of the IAC fits, not separate fits to the blue points; no additional reduced chi-squared is assigned to them.

The main-fit diagnostics accompany the figures. The following values cover the additional fit-window and all-scan alternatives quoted in the text. Scan-selection masks remain fixed when varying the fit window. Each rate-line variant uses rate standard deviations from a separate 300-replicate whole-scan bootstrap; the supplied delay and grating coordinates remain fixed. The same diagonal and unweighted-fit limitations apply, and uncertainties on these diagnostic values have not been estimated.

Additional fit	Diagonal χ^2_ν	ν
PP, 0.5–33 ps	0.141	19
PP, 0.5–57 ps	1.44	20
PP, 0.5–100 ps	1.89	21
PP, 0.5–2000 ps	120	59
PP, 1–2000 ps	129	54
PP, 2–2000 ps	11.1	52
TG slow-rate line, start 0.2 ps	1.32	2
TG slow-rate line, start 1 ps	2.46	2
TG slow-rate line, start 2 ps	0.168	2
TG slow-rate line, all scans retained	0.591	2

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AI Use Disclosure

Generative AI (OpenAI Codex) assisted with Python analysis, plotting, mathematical derivations, literature checks, and drafting and revising this report. Calculations and figures were generated from the supplied experimental files. AI assistance does not replace independent scientific review; responsibility for the submitted analysis and conclusions remains with the authors.